MILITARY SPECIFICATION

CONVERTER GROUP, "ENHANCED" DIGITAL SCAN OD-(___)/APQ-126(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification establishes the design, performance and test requirements for the "Enhanced" Digital Scan Converter Group (EDSCG) for use with the AN/APQ-126(V) Forward Looking Radar (FLR) in A-7D/E/K and TA-7C aircraft.

1.2 Classification - The EDSCG consists of the following items and should be identical to one of the following Indicator Groups, each differentiated by its peculiar Cathode Ray Tube Display (CRTD) complement.

1.2.1 Indicator Group OD-(__)/APQ-126(V), herein referred to as the A-7E EDSCG, consists of the following:

Converter, Signal Data	CV-3779/APQ-126(V) (SDC)
Indicator, Intratarget Data	IP-1418/APQ-126(V) (Forward CRTD)

1.2.2 Indicator Group OD-(__)/APQ-126(V), herein referred to as the TA-7C EDSCG, consists of the following:

Converter, Signal Data	CV-3779/APQ-126(V) (SDC)
Indicator, Intratarget Data	IP-1418/APQ-126(V) (Forward CRTD)
Indicator, Intratarget Data	IP-(__)/APQ-126(V) (Aft CRTD)

1.2.3 Indicator Group OD-186/APQ-126(V), herein referred to as the A-7D EDSCG, consists of the following:

Converter, Signal Data	CV-3779/APQ-126(V) (SDC)
Indicator, Intratarget Data	IP-1443/APQ-126(V) (Forward CRTD)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.4 Indicator Group OD-(__)/APQ-126(V), herein referred to as the A-7K EDSCG, consists of the following:

Converter, Signal Data	CV-3779/APQ-126(V) (SDC)
Indicator, Intratarget Data	IP-1443/APQ-126(V) (Forward CRTD)
Indicator, Intratarget Data	IP-(__)/APQ-126(V) (Aft CRTD)

1.3 Associated equipment - The indicator group should operate with the associated equipment listed in Table IV (See 6.3).

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards - Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-5088	Wiring, Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-R-10509	Resistors, Fixed Film (High Stabilty), General Specification for
MIL-E-19600	Electronic Modules, Aircraft, General Requirements for
MIL-M-38510	Microcircuits, General Specification for
MIL-C-38999	Connnector, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, and Breech Coupling), Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-R-55182	Resistors, Fixed Film, Established Reliability, General Specification for

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MIL-G-81132	Guidance Section, Guided Weapon MK39 MOD 2
MIL-F-81898	Forward Looking Radar Set, AN/APQ-126(V)

NAVAL AIR SYSTEM COMMAND

AS-1720	Converter Signal Conditioner

STANDARDS

FEDERAL

FED-STD-3	Colors, Aeronautical Lighting

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirement for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics, and Utilization of
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing
MIL-STD-965	Parts Control Program
MS25237-327	Lamps, Incandescent, Single Contact Midget Flanged Base (T-1-3/4 Bulb)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

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EIA-STD RS-343-A Electrical Performance Standards for High Resolution Monochrome Circuit Television Camera

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W., Washington, DC 20006.)

2.3 Order of Precedence - In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article inspection - This specification provides for first article inspection.

3.1.1 First article - When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3.1 and 6.4).

3.2 Parts and material - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

3.2.1 Parts selection - Standard parts (MS, AN, and MIL) shall be used to the maximum extent possible. Parts selection for electronic equipment shall be in accordance with MIL-STD-965. In the selection of electronic parts, preference shall be given to the use of TX and ER parts. Integrated Circuits not in accordance with MIL-M-38510 may be used provided the screening is equivalent to that required by MIL-M-38510. Parts which are not in accordance with MIL-M-38510 shall have their screening procedures reviewed concurrent with non-standard parts approval request. All standard parts shall be identified on drawings and data lists by the standard part number. Standard parts requiring qualification, but not covered by a government Qualified Products List (QPL) shall not be used.

3.2.2 Parts screening - All electronic, electrical, and electromechanical parts shall have been screened to special screening specifications; e.g., ER, TX, MIL-M-38510 or their equivalents, prior to installation into their respective assemblies. Where possible, candidate parts shall be selected from device types available to existing screening specification. Circuit matched "select-on-test" components shall not be used in the design of this equipment.

3.2.3 Magnetic materials - In the design of the CRTD's, the use of materials which easily become magnetized shall be held to a minimum consistent with the utilization of existing components.

3.2.4 Nonstandard parts and material approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistor, and diodes) other than microelectronic devices shall be obtained as outlined in MIL-E-5400 for the appropriate equipment classification specified in 3.3.

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3.3 Design and construction - The requirements of MIL-E-5400 shall apply to the design, construction, and workmanship with the exceptions and additions specified herein. The EDSCG equipment is classified as follows:

	EQUIPMENT	CLASS	CATEGORY	OPERATION
a.	Signal Data Converter (SDC)	2	III	Continuous
b.	Forward CRTD	1	III	Continuous and Intermittent
c.	Aft CRTD	1	III	Continuous and Intermittent

3.3.1 Total weight - Weight shall be a primary consideration in the selection of parts. The lightest practical design capable of the required performance shall be used. Maximum weight of the A-7D or A-7E EDSCG shall be not greater than 80 pounds (36.3 Kg). Maximum weight of the TA-7C or the A-7K EDSCG shall be not greater than 110 pounds (49.9 Kg).

3.3.2 Reliability - The contractor shall establish and maintain an effective reliability program in accordance with the requirements of MIL-STD-785. (See 4.4.2)

3.3.2.1 Operational stability - The EDSCG shall operate as required for a minimum of 100 hours in a temperature environment as specified herein, continuously or intermittently, without the necessity for adjustment of any controls which are inaccessible during flight.

3.3.2.2 Operating life - The EDSCG shall have as a minimum a service life of 4,000 hours when operated in any operational mode or combination of modes and under the environmental conditions specified herein with minimal service.

3.3.2.3 Storage - The equipment shall be designed to operate as required after 5 years storage and transit in accordance with MIL-STD-210. CRT degeneration is allowed after 18 months of continuous non-operating storage.

3.3.2.4 Reliability in mean time between failure (MTBF) - The TA-7C or A-7K EDSCG shall have a combined MTBF of 400 hours. The A-7D or A-7E EDSCG shall have a combined MTBF Of 500 hours.

3.3.2.5 Time totalizing meter - Each unit of the EDSCG shall contain a time totalizing meter in accordance with MIL-M-7793.

3.3.3 Cabling and connections - In addition to the following requirements for unit interface to the aircraft, the requirements of 3.3.7.3.3 shall also apply for purposes of test equipment interface. For internal connection, the requirements of 3.3.7.1.1 shall apply.

3.3.3.1 Connectors - All connectors of the EDSCG intended for unit level testing purposes or interconnection between units of the EDSCG or other aircraft systems shall be in accordance with MIL-C-38999. The design of con-

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nectors shall be such that reverse or improper assembly, mating, or installation is impossible.

3.3.3.2 Interconnecting cabling - The EDSCG shall be capable of required operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the EDSCG. External connectors with which the EDSCG Weapon Replaceable Assembly (WRA) connectors shall mate are listed in Table I. The EDSCG shall be electrically compatible with existing A-7D and A-7K aircraft wiring without change and shall be compatible with existing A-7E aircraft wiring except for deactivation of existing Auto-Tilt wiring. The EDSCG shall be compatible with existing TA-7C aircraft wiring without change except for the additional wiring required for the Auto-Tilt and Protection Interface capabilities.

3.3.4 Integral lighting - The embedded integral lighting of the EDSCG CRTD units shall be in accordance with MIL-P-7788 for Type V panels. Panels for A-7E and TA-7C CRTD units shall be Class 1-R; those for A-7D and A-7K CRTD units shall be Class 1-W.

3.3.5 Interchangeability - The EDSCG shall meet the interchangeability requirements of MIL-E-5400.

3.3.5.1 Retrofit interchangeability - The EDSCG units shall be physically interchangeable with other comparable units of the AN/APQ-126(V) FLR display subsystem in prior use as specified in Table II.

Operational interchangeability shall be as defined in 3.4.2.2.

3.3.5.2 EDSCG interchangeability - In addition to the requirements of MIL-E-5400, all contractor parts having the same part number shall be interchangeable with each other with respect to installation, performance, and testing. Parts interchangeability shall be in accordance with MIL-I-8500. Replacement of a WRA in the system (at Organizational level) shall be possible with no adjustment, alignment or harmonization of the replaced WRA or any other system component required (other than controls accessible to the equipment operator in flight). This capability shall be demonstrated on prototype models prior to production and on production units at the procuring activity's request.

3.3.6 Electromagnetic interference - The design of the EDSCG shall be such that its performance meets the EMI susceptibility and emission requirements of MIL-STD-461, Notice 3 for Class A1 equipment. Any equipment utilizing relays shall have suppression devices incorporated at the relay coils unless otherwise specified.

3.3.7 Maintainability - Contractor's maintainability efforts initiated during the predesign phase shall continue through the prototype phase. This continuing design liaison and guidance is performed to insure considerations of maintenance concepts, level of repair, maintainability requirement achievement and objectives. Procuring activity personnel will visit the contractor's facility as necessary to evaluate maintainability design conformance.

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3.3.7.1 Construction - In addition to the applicable requirements of MIL-STD-454, the requirements specified in 3.3.7.1.1 through 3.3.7.2.2 shall apply to the EDSCG.

3.3.7.1.1 Modular construction - Modular type construction shall be used to provide functionally oriented WRA's and SRA's (Shop Replaceable Assembly). The mechanical and electrical design features of the EDSCG pertaining to maintainability, accessibility, and interchangeabilty shall be in accordance with MIL-E-5400 and MIL-H-46855. Mechanical subassemblies shall be modularized to the extent practical to facilitate production assembly and rework. The use of cements and welds or other forms of permanent bonds which will consume appreciable rework time shall be avoided in the design of repairable assemblies. Assemblies that require machining, lapping, or custom fitting shall be avoided. Where two units must be matched and used with each other, an assembly drawing shall be created that identifies this requirement. When adjustment or alignments must be made, components shall be pinned, slotted, or scribed to permit rapid reassembly with minimum gauging and trial and error. The functional electronic circuitry or mechanical components shall be packaged into discrete replaceable SRA's. SRA's shall be designed as plug-in type modules with MIL-E-19600 as a guide. The design shall be such that reverse or improper assembly, mating or installation is impossible. Where possible, circuitry loops shall be contained in one SRA. In cases that the number of discrete components in a loop exceed the practical or physical space available on one SRA, mounting and packaging shall be in a manner most adaptable to maintainability. Non-repairable or disposable-on-failure subassemblies may be used and are preferred, providing such subassemblies are sufficiently reliable and inexpensive to justify that concept, or where significant weight savings may be achieved or where mechanical integration with or modification of existing design requires encapsulated subassemblies.

3.3.7.1.2 Encapsulation - Repairable SRA's/Assemblies shall not be encapsulated, potted, or embedded. When a conformal coating is used, it shall be a transparent type which is easy to remove and reapply.

3.3.7.2 Remove/Replace provisions - Design of the system shall make provisions for rapid removal and replacement of each WRA in the system with no requirement to remove or displace other WRA's/equipment. The design of the WRA shall continue this philosophy in that provisions shall be made for the rapid removal and replacement of SRA's and other replaceable items without removal of other parts, subassemblies, etc. The CRT, deflection yoke, optics assembly, and front panel controls of the CRTDs are excepted from this requirement as are the line filter and carrier board of the SDC.

3.3.7.2.1 Fasteners - The different types of fasteners (screws, bolts, quick release fasteners, etc.) used in the system shall be minimized to reduce tool requirements for maintenance.

3.3.7.2.2 Standard, special tools and test equipment - The contractor's equipment shall be designed so that standard tools and test equipment may be used to the greatest possible extent to accomplish all necessary adjustments, maintenance and testing.

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3.3.7.3 Maintainability implementation - The design, configuration, installation and operation of the equipment shall minimize maintenance/support requirements, including the need for special tools and support equipment, personnel skills, manpower, elapsed maintenance time and total maintenance time at all levels of maintainance. The design shall permit accomplishment of the maximum number of tasks at Intermediate level with all maintenance tasks capable of being performed either at the Organizational, Intermediate or Depot Levels (See 6.5.2). Design and construction provisions and allowance for maintenance/support of MIL-E-5400 shall apply.

3.3.7.3.1 Maintainability requirements - The Intermediate level elapsed maintenance manhours mean-time-to-repair (MCT) shall be not greater than 2.5 hours in case of failure of the EDSCG. The maximum corrective maintenance time (Mmaxct) shall be not greater than 4.5 hours at the 95th percentile value. The intermediate corrective maintenance actions include all elements of maintenance. If the discrepant SRA is repairable at Intermediate level, the time to fault isolate, repair and test the SRA shall be included in the Intermediate level meantime-to-repair. Equipment shall be operated on a "run to failure" concept and shall require no scheduled maintenance (See 6.5.1). Repair times shall be computed on the basis of accomplishment by maintenance personnel equivalent to Navy "A" school graduates. All Organizational level maintenance shall be capable of being performed under severe meteorological environments in which the aircraft is expected to operate from, using only common hand tools.

3.3.7.3.2 Organizational level test provisions - Organizational level testing shall be by built-in-test (BIT).

3.3.7.3.3 Intermediate and depot test provisions - The equipment shall be designed for compatibility with automatic checkout equipment and shall include test provisions to permit diagnostic and performance verification testing at Intermediate and Depot Level maintenance. Each WRA shall have sufficient accessible external test points that, when used in conjunction with the operating connector, will permit determination that performance of the WRA is within all specified parameters and tolerances or if substandard performance is determined, permit fault isolation to the SRA as specified in (a) through (d) of this paragraph. Where multiple test points are required, they shall be grouped into multipin connectors for ease of attaching test equipment. Test points shall, when used in conjunction with operational connectors, provide the means for:

a. Performance of an overall or end-to-end quantitative check.

b. Measuring the necessary input/output parameters to enable isolation of any and all faulty SRA's of the WRA to the following criteria:

(1) In at least 90 percent of the cases of probable malfunction of an SRA, the fault shall be isolated to the sole SRA.

(2) In 95 percent, or more, of the cases of probable malfunction of an SRA, the fault shall be isolated to that SRA and no more than one other SRA.

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(3) In all cases of probable malfunction of an SRA, the fault shall be isolated to that SRA and no more than two other SRA's.

c. Measurement of necessary input/output parameters to enable calibration or alignment of the WRA.

d. Introducing any other necessary functional inputs and dummy loads necessary for test.

3.3.7.3.3.1 Support equipment interface requirements - Each EDSCG WRA and each EDSCG SRA designated in Table III shall be equipped with identification resistors which shall provide a unique passive electrical identification of the UUT and the UUT modification series. The signature circuitry shall consist of two MIL-R-10509 ± 1 percent (except as superseded by MIL-R-55182) tolerance resistors selected from alternate values starting with 90.0 ohms and extending to 1.1 megohms. Assigned resistor values for each WRA and SRA are included in Table III. For SRA's Resistor No. 1 shall be connected to operational connector contacts numbers 4 and 5; Resistor No. 2 shall be connected to operational connector contacts numbers 5 and 6. For connectors whose contacts are designated by alphabetic characters only, contacts D, E, and F shall be used in lieu of 4, 5, and 6 respectively. In the case of a WRA, the contractor shall assign operational connector contacts for the signature resistors. Identification resistors connected to test points will only be acceptable when:

a. Input/output requirements eliminate any possibility of connecting them to the operational connectors.

b. Test points are readily accessible for quick connection to the test equipment,

Common connectors (identical part numbers) shall be used in the design of submodules where feasible, thus reducing the number of test receptacles and adapter cables for Intermediate level and Depot level support equipment. Function assignments to connector pins shall be standardized where practical.

3.3.7.3.3.2 Shop repair test points - Shop repair test points are required on each SRA. Shop repair test points are used to quantitatively determine the performance level of the SRA and, in the event of a failure, to diagnose which non-repairable, or small group of non-repairables, within the SRA is at fault. All SRA's shall contain adequately defined test points such that, in conjunction with the input/output connector, they provide the means for the measurement of necessary input/output parameters to enable calibration or alignment of the SRA. Test Points shall be designed such that damage to a Unit Under Test (UUT) shall not result from a short circuit between any test point and ground.

a. For SRA's newly designed for the EDSCG, the following test point requirements shall apply:

(1) Where multiple test points are required, they shall be grouped into multipin connections for ease of attaching the equipment. SRA test connectors shall be a common type and shall not be keyed.

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(2) The shop repair test points in conjunction with the input/output connector shall provide the means for measuring the necessary input/output and circuit parameters to enable isolation of the faulty non-repairables as follows:

(a) When the SRA contains 10 or fewer nonrepairables, isolation to groups of 2 or less shall be possible for 50 percent of the possible faults, Isolation to 4 or less must be possible for all possible faults.

(b) When the SRA contains more than 10 nonrepairables, isolation to groups of 4 or less shall be possible for 80 percent of the possible faults. Isolation to groups of 8 or less must be possible for 95 percent of the possible faults. Isolation to groups of 10 or less must be possible for 100 percent of the possible faults.

b. Existing SRA hardware designs that are a part of the design of different models of like equipment may be incorporated into the EDSCG design in accordance with the following relaxed test point requirements:

(1) The shop repair test points in conjunction with the input/output connector shall provide the means for measuring the necessary input/ output and circuit parameters to enable isolation of the faulty nonrepairables as follows:

(2) When the SRA contains more than 10 non-repairables, isolation to groups of 6 or less shall be possible for 50 percent of the possible faults. Isolation to groups of 12 or less must be possible for 90 percent of the possible faults.

3.3.7.4 Built-in-test equipment (BITE) - The EDSCG shall include built-in-test provisions for preflight and troubleshooting purposes. The self-test feature shall include provisions for monitoring radar system inputs and providing FAIL/SAFE outputs to the Radar Fault Locator Unit. In addition, the BIT provisions shall be capable of isolating 95 percent of all failures within the EDSCG to the faulty WRA by Fault Locator indications and CRTD display indications, Operational status of the avionic system is determined using BITE to perform a set of functional tests that are necessary to assure the operational usability of the avionics system or avionic set. The BIT features shall be such that the need for additional test equipment at Organizational level is not required.

3.3.7.4.1 AGR/TF self-test capability - Provisions shall be made to verify AGR lock-on/track operation of the AGR processor within the SDC. A capability to verify the SDC-to-NWDC AGR data channel via BIT shall also be included. Both AGR and TF self-test provisions shall be incorporated into the SDC design as specified in 3.4.8.3.

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3.3.7.4.2 Fail-safe features - All system fault isolation and self test circuitry shall be designed to exhibit fail-safe features. The fail-safe features shall be understood to mean that faults occurring in any part of the test circuitry shall not induce a failure in any other system(s) circuitry nor in the circuitry of the associated equipment listed in Table IV. (See 6.3)

3.3.8 Identification product - A standard nameplate in accordance with MIL-STD-130 shall be permanently attached to each EDSCG WRA. Parts not requiring nameplates shall be marked for identification in accordance with MIL-STD-130.

3.3.9 Standard conditions - The following conditions shall be used to establish normal performance characteristics and for making laboratory bench tests:

a.	Temperature	Room ambient (25^0 C $\pm$ 10^0 C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power Voltage	115 $\pm$ 1 VAC - 400 Hz 27.5 $\pm$ 0.5 VDC

3.3.10 Service conditions - The EDSCG shall be designed to meet the aircraft installed requirements of this specification when exposed to the following environmental conditions or reasonable combinations of these conditions as specified in MIL-E-5400 for the appropriate equipment classification specified in 3.3, except as modified herein:

3.3.10.1 Temperature/Altitude (operating)

a. SDC:

Continuous operation from -54^0 C to $+71^0$ C from sea level to 50,000 feet as bounded by Figure 6, Sheet 2, Curve A of MIL-E-5400 for Class 2 equipment.

b. CRTD's:

Continuous operation from -54^0 C to $+55^0$ C from sea level to 50,000 feet as bounded by Figure 6, Sheet 1, Curve A of MIL-E-5400 for Class 1 equipment. Intermittent operation (up to 30 minutes) up to $+71^0$ C from sea level to 50,000 feet as bounded by Figure 6, Sheet 1, Curve B of MIL-E-5400 for Class 1 equipment.

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3.3.10.2 Temperature (non-operating)

a. SDC:

As required by MIL-E-5400 for Class 2 equipment

b. CRTD's:

As required by MIL-E-5400 for Class 1 equipment

3.3.10.3 Humidity - Zero to 100 percent relative humidity at temperatures from $+36^{0}$ C to $+55^{0}$ C. (See 4.3.2.3)

3.3.10.4 Vibration - (See 4.3.2.4)

a. SDC:

Figures 1, 2, 3, 4, 5, and 6 of this specification shall apply.

b. CRTD's:

Figures 7, 8, 9, and 10 of this specification shall apply.

3.3.10.5 Shock - The EDSCG shall be capable of withstanding 11 millisecond shock loads as follows: (See 4.3.2.5)

a. Operational:

20g lateral and longitudinal, 32g, 23g, and 25g vertical on Forward CRTD, Aft CRTD and SDC respectively.

b. Crash Safety (Mounting Base)

40g except that the Forward CRTD must withstand 42g vertical.

3.3.10.6 Sand and dust - The EDSCG shall withstand, in both the operating and non-operating conditions, exposure to sand and dust particles as encountered in desert areas. (See 4.3.2.6)

3.3.10.7 Fungus - The EDSCG shall withstand, in both the operating and non-operating condition, exposure to fungus growth as encountered in tropical climates. (See 4.3.2.7)

3.3.10.8 Salt atmosphere - The EDSCG shall withstand, in both the operating and nonoperating conditions, exposure to a salt sea atmosphere. (See 4.3.2.8)

3.3.10.9 Explosive conditions - The EDSCG shall not cause ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere. (See 4.3.2.9)

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3.3.11 Input electrical power - The EDSCG shall meet the requirements of MIL-STD-704 for Category B equipment and shall be capable of providing specified performance from power sources with characteristics as defined in MIL-STD-704 with the exception that degraded performance is allowable for 500 milliseconds after initiation of the transients as specified in MIL-STD-704. Power requirements shall be updated during the production program. The power input shall be as follows:

a. 115 VAC, 3Ø, 400 Hz, 0.8 PF MIN, 3 Amperes per phase

b. 28 VDC, 2.5 Amperes

c. 5 VAC (Lighting), 2 Amperes

The EDSCG shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within normal limits.

3.3.12 Cooling - An exhaust or recirculation fan, mounted integral with each unit of the EDSCG, shall be used to provide cooling of that unit.

3.3.12.1 SDC cooling air - Conditioned air with flow rate and temperature characteristics as defined in Table V shall be supplied through ducts which are adjacent to but not connected to the cooling air inlet on the SDC. With the air conditioner inoperative, compartment ambient air shall be supplied. The compartment ambient in flight shall be within the MIL-E-5400, Class 2 thermal limit. During the Emergency Operation, i.e., when the air conditioning system is inoperative, the equipment shall continue to operate as required up to a maximum compartment temperature equivalent to MIL-E-5400, Class 2, Curve A.

3.3.12.2 CRTD cooling air - Air supplied for CRTD cooling shall not be specially conditioned.

3.3.13 Safety - The EDSCG equipment shall be designed for safety in handling, operating and servicing with particular attention being paid to eliminating hazards of radiation, shock and burns as specified in MIL-STD-454. Controls shall be designed and located to preclude inadvertant operation.

3.4 Performance - The EDSCG performance shall be compatible with the flight characteristics of the aircraft and operating characteristics of the associated equipment listed in Table IV. (See 6.3) Detailed performance characteristics of the EDSCG shall be as specified herein, exclusive of errors in input signals. A matrix summarizing display features and major EDSCG functions by mode is provided in Table VI.

3.4.1 Operational capabilities - The EDSCG shall provide for the following operational capabilities:

a. PPI Display

(1) Ground Map Pencil (GMP) and Ground Map Shaped (GMS) (ground range corrected)

(2) Terrain Avoidance (TA) (slant range)

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(3) Beacon (BCN); Airborne BCN (slant range) and Ground BCN (ground range corrected)

(Airborne BCN mode shall be enabled by the presence of the BCN command in the absence of the CDCE Operate command. Ground BCN shall be enabled by the presence of the BCN command in conjunction with the CDCE Operate command.)

(4) Cross scan (CS) display (reduced scan, azimuth display only); TA (slant range) and GMP (ground range corrected)

b. Expanded Display (B scan, slant range)

c. Frozen Map Display (HOLD mode)

d. Terrain Following (TF) Display (E2 presentation)

e. Air-to-Ground Ranging (AGR); Clutter/point target detection/tracking and Display

f. Shrike Improved Display System (SIDS) Display

g. Television (TV) Display (525 line)

h. Forward Looking Infrared (FLIR) Display (875 line)

i. Navigation System Alarm Monitoring

j. Radar System and EDSCG Fault Isolation

k. Radar Silence selection in HOLD, TV, FLIR, and SIDS modes

l. Auto Antenna Tilt selection

3.4.2 Compatibility - The EDSCG shall be compatible with the remainder of the AN/APQ-126 FLR system and the requirements of the aircraft/avionics equipment as follows:

3.4.2.1 Mode compatibility - The EDSCG shall be compatible with the Beacon Aided Radar Bomb (BARB) and EXPAND mode modifications present in an "Enhanced" AN/APQ-126 FLR.

3.4.2.2 FLR Power Supply - Programmer (PSP) compatibility - No damage shall be incurred by either the EDSCG or the remainder of the AN/APQ-126 FLR system when the FLR system is operated while configured with PP-6130/APQ-126(V). In this system configuration some of the EDSCG capabilities listed herein shall not function. With PP-7799/APQ-126(V) (an Enhanced PSP), an EDSCG is required for any operation of the AN/APQ-126 FLR system.

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3.4.2.3 Aft CRTD compatibility - All SDC units procured shall be capable of operating properly in the dual indictor A-7K or TA-7C EDSCG subsystem configuration which requires simultaneous operation of both the Forward CRTD and the Aft CRTD. Unless otherwise specified herein, statements referring to the CRTD(s) in this specification shall apply to the Forward CRTD only in the A-7D/E EDSCG configuration, but shall apply equally to both CRTD's in the A-7K/TA-7C EDSCG configurations.

3.4.3 Display requirements - The required performance for each mode shall be as indicated in the following paragraphs.

3.4.3.1 CRT characteristics - The CRTD shall employ a high brightness, high resolution CRT to display FLR and other sensor information. All of the CRT faceplate area contained within the 4 inch (10.16 cm) circular viewing area of the CRTD shall be usable display area with respect to the CRT requirements specified in the following paragraphs.

3.4.3.1.1 CRT brightness requirements - When operated in a TV display format as specified in 3.4.3.8, the CRT maximum displayed brightness shall be a minimum of 1000 foot-lamberts (3426 candelas/m) measured at the CRT face exclusive of any filters. The display brightness, measured under the following conditions with a 0.10 inch (0.25 cm) photometer spot size on the face of the CRT, shall not vary more than ±20 percent over the entire usable display area:

a. Brightness - 1000 foot-lamberts (3426 candelas/m), minimum

b. CRT spot size - 0.005 inch (0.013 cm) to 0.007 inch (0.018 cm)

c. Raster scan - 2:1 interlace raster at a 30 Hz rate

d. Raster lines - 525 lines (512 active lines)

e. Raster size - 4 inch x 4 inch (10.16 cm x 10.16 cm)

3.4.3.1.2 CRT resolution requirements - When operated in a TV display format as specified in 3.4,3.8, the CRT shall have a resolution of no less than 600 horizontal elements for a 700 element input and no less than 3 shades of gray contrast under cockpit ambient lighting conditions of 10,000 foot-lamberts (34,260 candelas/m). Under ambient cockpit lighting of 1000 foot-lamberts (3426 candelas/m2) or less, a minimum of 8 shades of gray with filter installed shall be available. The CRT shall have a resolution of at least 185 shrinking raster lines per inch (73 shrinking raster lines per cm) with 1000 foot-lamberts (3426 candelas/m2) brightness at the face of the CRT.

3.4.3.1.3 CRT phosphor/faceplate specifications - Non-uniformities in the phosphor coating or faceplate glass of the CRT shall be acceptable to the degree specified in the following paragraphs.

3.4.3.1.3.1 Spot defects - Phosphor spot defect acceptability criteria are as follows:

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a. Criteria for non-elongated spots - Spots which are not elongated and are less than the following dimensions, will be accepted:

Type of Spot	Inch	Cm
Dead	0.025	0.064
Bright	0.025	0.064
Color	0.050	0.127

b. Criteria for elongated spots

(1) Elongated spots of any length whose maximum widths are 0.010 inch (0.025 cm) or less will be accepted.

(2) Elongated spots between 0.010 inch (0.025 cm) and 0.020 inch (0.051 cm) in width whose length is 0.500 inch (1.27 cm) or less shall be included in the assessment of the total number of spots in accordance with Table VII. A CRT shall be considered defective if it has elongated spots between 0.010 inch (0.025 cm) and 0.020 inch (0.051 cm) in width and more than 0.500 inch (1.27 cm) in length.

(3) The diameter of elongated spots whose width is more than 0.020 inch (0.051 cm) shall be taken as half the sum of the length and width. The resultant diameter shall be assessed in accordance with Table VII.

3.4.3.1.3.2 Scratches - Faceplate scratch acceptability criteria are as follows:

a. A scratch of any length whose width is less than 0.0015 inch (0.0038 cm) shall be accepted.

b. A CRT having scratches of 0.0015 inch (0.0038 cm) to 0.003 inch (0.008 cm) in width whose combined total length is more than 1 inch (2.54 cm) shall be considered defective.

c. A CRT having scratches of 0.003 inch (0.008 cm) to 0.005 inch (0.013 cm) in width whose combined total length is more than 0.25 inch (0.64 cm) shall be considered defective.

d. A CRT having any scratches whose width is more than 0.005 inch (0.013 cm) shall be considered defective.

3.4.3.1.3.3 Blemishes (inside and outside surface blemishes) - All degrees shall be accepted which are not visible to the unaided eye when viewed along the axis of the CRT from a distance of three times the screen diameter or diagonal, or 12 inches (30.5 cm), whichever is greater. Visible blemishes shall be classified as dead spots and shall be in accordance with the requirements specified in Table VII.

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3.4.3.2 PPI display - The PPI display format shall be a depressed center, drift stabilized, spiral scan raster covering a 90 degree circular sector. The spiral scan raster shall be composed of two positive 2:1 interlaced fields of 256 active lines each to provide 512 active lines in a 525 line frame. The field rate shall be 60.0 ± 0.1 Hz. The CRTD shall present a minimum of 512 discrete range bins in accordance with a display height of 3.95 inches (10.03 cm) and a minimum of 160 discrete azimuth bins in accordance with a display width of 90 (1.57 rad). The CRT maximum displayed brightness shall be a minimum of 1000 footlamberts (3426 candelas/m2) measured at the CRT face exclusive of any filters. The PPI format shall be capable of displaying 8 (square root of 2) shades of gray.

3.4.3.2.1 PPI display accuracy - The PPI modes shall provide the following accuracies, exclusive of any input signal errors:

a. Raster apex - The raster apex shall be positioned within the bottom 1/16 inch (0.159 cm) of the viewing area, centered horizontally ± 1/16 inch (0.159 cm).

b. Raster amplitude - The raster vertical height shall be such that the raster line corresponding to the value of the selected range scale shall always lie within the top 1/16 inch (0.159 cm) of the display viewing area.

c. Video range registration accuracy - The EDSCG shall have the capability to place the radar video on the proper raster line to an accuracy of -2, +1 raster lines. Each raster line shall represent the increment of range specified below for each applicable selected range scale:

Selected Range Scale	Range Line Increment
(1) 5 nm (9.26 km)	60.00 feet (18.29 m) ± 0.03 foot (0.9 cm)
(2) 10 nm (18.52 km)	120.00 feet (36.58 m) ± 0.06 foot (1.8 cm)
(3) 20 nm (37.04 km)	240.00 feet (73.15 m) ± 0.12 foot (3.7 cm)
(4) 40 nm (74.08 km)	480.00 feet (146.30 m) ± 0.24 foot (7.3 cm)
(5) 80 nm (148.16 km)	960.00 feet (292.61 m) ± 0.48 foot (14.6 cm)

d. Video azimuth registration accuracy - The angular placement of targets on the display shall represent the true azimuth position of the targets, as presented by inputs from the radar, to an accuracy of ± 0.5625 degree (9.817 mrad) which is defined to be one PPI azimuth bin. Azimuth motion of scintillation-free targets shall be no greater than one azimuth bin.

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e. Manual range cursor accuracy - The EDSCG shall have the capability to place the manual range cursor on the appropriate raster line to an accuracy of ± 1 raster line. Cursor movement (jitter) shall be no greater than 1 raster line.

f. Driven range cursor accuracy - The EDSCG shall have the capability to place the driven range cursor on the proper raster line to an accuracy of ± 1 raster line.

g. Driven azimuth cursor accuracy - The EDSCG shall have the capability of placing the azimuth cursor in the proper azimuth location relative to the azimuth position signal input to an accuracy of ± 0.5625 degree (9.817 mrad).

3.4.3.2.2 PPI range scales - The following Range/Mode combinations shall be provided in PPI display modes with the following specific range coverage. Range selection shall be accomplished by sequencing from maximum to minimum range upon an external switch closure as specified in 3.4.4.

a. GMP/GMS, BCN

(1) 5 NM (9.26 km) 30,720 foot (9363.5 m) total coverage

(2) 10 NM (18.52 km) 61,440 foot (18,727 m) total coverage

(3) 20 NM (37.04 km) 122,880 foot (37,454 m) total coverage

(4) 40 NM (74.08 km) 245,760 foot (74,908 m) total coverage

(5) 80 NM (148.16 km) 491,520 foot (149,815 m) total coverage

b. TA-CSTA, CSGMP

(1) 5 NM (9.26 km) 30,720 foot (9363.5 m) total coverage

(2) 10 NM (18.52 km) 61,440 foot (18,727 m) total coverage

3.4.3.2.3 PPI ground range correction - In GMP, CSGMP, GMS, and Ground BCN modes, the display shall be corrected to simulate a true ground range map. The altitude used for this correction computation shall be as commanded by the Navigation Weapon Delivery Computer (NWDC) via the CDCE data channel as specified herein. The digital channel altitude data shall be used to determine the appropriate delay and slope of the correction factor applied. Upon receipt of a NWDC Fail discrete or an invalid condition of the digital data channel, the altitude used shall be set to zero feet by the SDC. The accuracy of the ground map shall be as specified herein.

3.4.3.2.4 PPI display symbology - Symbology as specified in the following paragraphs shall be present on the display in the modes specified.

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3.4.3.2.4.1 Manual range cursor - A manual range cursor shall be presented in GMP and GMS mode in the absence of a CDCE operate command from the radar set control as well as in CSGMP and Air BCN modes. The position of the cursor shall be controlled by the BCD strobe pulse from the radar range set control. A 5 MHz clock and Maximum Range Gate pulse shall be provided to enable generation of the BCD strobe. The accuracy of the cursor placement shall be as specified in 3.4.3.2.1. The intensity of the cursor shall be adjustable as specified in 3.5.2.4.4 and 3.5.3.4.4.

3.4.3.2.4.2 Driven cursors - The EDSCG shall have the capability to display range and azimuth cursors upon receipt of the CDCE Operate discrete in conjunction with selection of GMP, GMS or Ground BCN mode. The position of the cursors shall be as commanded by the NWDC via the data channel specified in 3.4.6.3. The accuracy of the cursor placement shall be as specified in 3.4.3.2.1. The intensity of the cursors shall be adjustable as specified in 3.5.2.4.4 and 3.5.3.4.4.

3.4.3.2.4.3 PPI range readout - When driven cursors are commanded, the EDSCG shall write the slant range in nautical miles (rounded to tenths of a mile) to the range cursor in the upper right hand portion of the CRT of the CRTD. The cursor intensity control as specified in 3.5.2.4.4 and 3.5.3.4.4 shall be used to control the intensity of the range readout.

3.4.3.3 Expanded display - A switch shall be provided on the CRTD to enable selection of the EXPAND mode. The expanded display may be entered whenever the EDSCG is in a NWDC controlled (CDCE operate command present) mapping or Ground BCN mode with the range cursor commanded to less than 20.0 $\pm$ 0.5 nautical miles (37.0 $\pm$ 0.9 km) and the azimuth cursor commanded to less than $\pm$ 45 ($\pm$ 0.79 rad). If the EXPAND mode is selected via this switch, when the conditions are not met, the display shall remain as previously selected. Also, whenever the required conditions are exceeded while in EXPAND mode, the EDSCG shall revert to the PPI mode. Upon occurrence of a NWDC Fail discrete, entry into EXPAND shall be inhibited; if the EDSCG is operating in the EXPAND mode, it shall revert to PPI mode. Upon exiting the EXPAND mode, the SDC memory shall be automatically cleared before storing fresh data. The expanded display format shall be a B-scan raster composed of two positive 2:1 interlaced fields of 256 active lines each to provide 512 active lines in a 525 line frame. The field rate shall be 60.0 $\pm$ 0.1 Hz. The display shall be centered in range about the commanded range cursor unless the range cursor is commanded to less than 7,680 feet (2341 m) by the NWDC; then the display shall remain centered about 7,680 feet (2341 m) slant range. In azimuth, the display shall be centered about the null point of the antenna azimuth position signal derived from the azimuth resolver of the radar. The azimuth coverage shall be equal to $\pm$ 20 (0.349 rad), with the scan utilizing the full display except when the pointing command plus drift exceeds $\pm$ 35 (0.611 rad), Azimuth memory elements not addressed on a scan-to-scan basis shall be blanked. The expanded display format shall be capable of displaying 8 (square root of two) gray shades.

3.4.3.3.1 Expanded display (B-Scan) accuracies - The expanded mode shall provide the following range and bearing accuracies exclusive of input signal errors:

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a. Raster size and centering - The raster horizontal and vertical amplitude shall be 3-15/16 ± 1/16 inches (10.00 ± 0.16 cm). The raster shall be centered ± 1/16 inch (± 0.16 cm) horizontally and vertically.

b. Video range registration accuracy - The B-scan display shall display over a range from 7,680 feet (2341 m) in front of the commanded cursor position to 7,680 feet (2341 m) beyond the commanded cursor position to an accuracy of ± 2 range lines. The display range resolution shall be 30.00 ± 0.02 feet (9.14 m ± 0.5 cm) for each raster line. Video positioning accuracy with respect to the displayed range cursor shall be defined as the following:

Range to Cursor, (R_C) < 30K ft. (9.14 km),

Accuracy = ± 2 range lines

R_C ≥ 30K ft. (9.14 km),

Accuracy = ± integer value of

$$R_C\left(\frac{2.5 \times 10^{-3}}{30}\right) + 0.5$$

expressed in range lines

c. Video azimuth registration accuracy - The EDSCG shall have the capability to measure the synchro resolver antenna position data and place the corresponding video on the display, relative to the centered cursor, to an accuracy of 0.250 degree (4.36 mrad) which is defined to be one EXPAND azimuth bin.

d. Range cursor accuracy - For range cursor locations of less than 7680 feet (2341 m), the cursor shall be placed on the proper raster line to an accuracy of ± 2 raster lines.

3.4.3.3.2 Expand range scale - In the EXPAND mode, the range displayed shall be 15,360 feet (4681.7 m), independent of range scale selected.

3.4.3.3.3 Expand display symbology - Symbology as specified in the following paragraphs shall be present on the display as specified.

3.4.3.3.3.1 Driven range cursor - The EDSCG shall have the capability to display a range cursor whenever operating in EXPAND mode. The cursor shall be positioned on the 256th raster line, unless it is commanded to less than 7680 feet (2341 m). In the latter case, the range cursor shall be positioned as commanded by the NWDC via the data channel specified in 3.4.6.3 and referenced to the bottommost raster line defined to be zero range. The accuracy of the cursor placement shall be as specified in 3.4.3.2.1. The intensity of the cursor shall be adjustable as specified in 3.5.2.4.4 and 3.5.3.4.4.

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3.4.3.3.3.2 Driven azimuth cursor - The EDSCG shall have the capability to display an azimuth cursor whenever operating in EXPAND mode. The cursor shall be positioned in the center of the display, regardless of the azimuth cursor position commanded by the NWDC. The intensity of the cursor shall be adjustable as specified in 3.5.2.4.4 and 3.5.3.4.4.

3.4.3.3.3.3 Expand range readout - The EDSCG shall write the slant range in nautical miles (rounded to tenths of a mile) to the range cursor in the upper right hand portion of the display. The cursor intensity control as specified in 3.5.2.4.4 and 3.5.3.4.4 shall be used to control the intensity of the range readout.

3.4.3.3.4 Expand mode discrete - The EDSCG shall command EXPAND mode by issuing a 28V discrete to the FLR PSP to select antenna pointing angle data. Neither entry into nor exit from EXPAND mode shall result in a condition of antenna scan such that the antenna azimuth motor attempts to drive the antenna to a position beyond its mechanical stop points.

3.4.3.4 Frozen map display (HOLD mode) - The SDC memory shall provide data storage sufficient to contain a complete display frame of radar video return information. The EDSCG shall have the capability to interrupt the updating of radar video return information in GMP, CSGMP, GMS, Ground BCN, and all EXPAND modes. The HOLD mode shall be selectable by a switch on the front panel of the CRTD. Whenever operating in HOLD mode, the only means of clearing the SDC memory shall be by exit from HOLD mode, entry into SIDS mode, or selection of a FLR mode for which HOLD is invalid. A HOLD advisory lamp shall be provided on the CRTD which shall be illuminated only when the HOLD mode has been selected and is legal for the radar operating mode. The HOLD lamp shall not illuminate in radar modes other than GMP, CSGMP, GMS, Ground BCN, and EXPAND. The HOLD lamp shall illuminate in TV and FLIR when HOLD is selected and is legal for the operating radar mode. The HOLD advisory lamp shall be dimmed from the cockpit dimming control in conjunction with the range scale advisory lamps; therefore, separate isolated HOLD advisory lamp excitation signals shall be provided by the SDC to prevent crosstalk in the dual-indicator TA-7C and A-7K EDSCG configurations. The signal interconnect for the HOLD advisory lamp is listed in Tables VII, IX, and X.

3.4.3.4.1 Hold display symbology - While operating in PPI HOLD mode, the range readout and any cursors present on the display shall be moveable with respect to the "frozen" radar return video. Altitude information used to ground range correct the range cursor position shall be frozen in PPI HOLD Mode. While operating in EXPAND HOLD mode, all display symbology (both readout value and cursor position) shall be "frozen".

3.4.3.4.2 Hold mode operation in dual indicator EDSCG's - With the EDSCG installed in the A-7K or TA-7C aircraft, HOLD shall be selected only from the controlling cockpit. The "28V Switched" signal shall be provided to the CRTD in only one cockpit; presence of this signal shall serve to designate the controlling cockpit. The HOLD advisory lamp shall illuminate on both displays when HOLD is selected and is legal for the radar operating mode. Both HOLD lamp discrete outputs shall be diode isolated in the SDC to prevent improper operation of the cockpit dimming function.

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3.4.3.5 Terrain following (TF) display - The TF display format shall be an E-scan raster denoting elevation in the vertical axis and range (nonlinear) in the horizontal axis. The antenna elevation angle shall be a linear representation in the vertical dimension and the elevation coverage shall be from +7 degrees (0.12 rad) to -18 degrees (0.31 rad). The TF raster shall be composed of two positive 2:1 interlaced fields of 256 active vertical raster lines to provide 512 active vertical raster lines for each frame. The field rate shall be 60.0 ± 0.1 Hz. The CRTD shall present a minimum of 512 discrete range bins comforming to a displayed width of 3.00 inches (7.62 cm) and a minimum of 80 discrete elevation bins for a displayed height of 2.25 inches (5.72 cm). The CRT maximum displayed brightness shall be a minimum of 1000 foot-lamberts (3426 candelas/m^2) measured at the CRT face exclusive of any filters. The TF format shall display 2 gray shades representing level 0 (no video) and level 2 or greater.

3.4.3.5.1 TF display accuracy - The TF display shall have the following accuracies:

a. Raster size and centering - The TF display raster vertical amplitude shall be 2.25 ± 1/16 inches (5.72 ± 0.16 cm) and the horizontal amplitude shall be 3.00 ± 1/16 inches (7.62 ± 0.16 cm). The raster shall be centered on the display ± 1/8 inch (± 0.32 cm) vertically and horizontally.

b. Video range registration accuracy - The EDSCG shall have the capability to place the radar video on the proper raster line to an accuracy of ± 0.1 NM (± 0.2 km) or ± 10 percent of range to target whichever is greater.

3.4.3.5.2 TF range scale - The range displayed shall be 61,440 feet (18.73 km). Each raster line shall represent the appropriate incremental range bin size to provide an approximately logarithmically scaled range presentation.

3.4.3.6 AGR display - The AGR display format shall be a B-scan raster composed of two positive 2:1 interlaced fields of 256 active lines each to provide 512 active lines in a 525 line frame. The field rate shall be 60.0 ± 0.1 Hz.

3.4.3.6.1 AGR display accuracy - The AGR display shall exhibit the following accuracies:

a. Raster size and centering - The AGR display raster horizontal and vertical amplitude shall be 3-15/16 ± 1/16 inches (10,00 ± 0,16 cm), The raster shall be centered ± 1/16 inch (0.16 cm) horizontally and vertically.

b. Range strobe registration accuracy - The EDSCG shall have the capability to place the range strobe on the proper raster line to an accuracy of ± 2 percent of maximum range.

3.4.3.6.2 AGR range scale - The range displayed shall be 61,440 feet (18.73 km). Each raster line shall represent a nominal range increment of 120 feet (36.6 m).

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3.4.3.6.3 AGR display symbology - Symbology as specified in the following paragraphs shall constitute the display as specified.

3.4.3.6.3.1 AGR strobe and search line symbol - The AGR strobe, which denotes the range to the track gate (range to target in track mode), shall be a horizontal line segment 0.5 ± 0.1 inch (1.3 ± 0.3 cm) long centered horizontally on the search line symbol. The search line symbol shall be a vertical line centered horizontally on the display ± 1/16 inch (± 0.16 cm).

3.4.3.6.3.2 AGR range readout - In the AGR mode, the EDSCG shall write the slant range, in thousands of feet (rounded to the nearest tenth of thousand) to the center of the track gate, in the upper right hand portion of the CRT. The cursor intensity control shall be used to control the intensity of the range readout.

3.4.3.7 SIDS display - The EDSCG shall have the capability to display azimuth and elevation steering information in response to data provided from the SIDS signal converter CV-2529A/APG. The SIDS display function shall be compatible with the requirements of AS-1720 with the additional requirements specified in "a." through "c." below:

a. Write command characteristics - This SIDS Write Command signal will exist nominally at either 0 VDC or 12 VDC. Less than 0.5 VDC shall command an absence of video on the display, A nominal signal level of 12 VDC shall cause the CRT to display a video target at the position represented by the azimuth and elevation deflection commands. Time duration for the write command will be a minimum of 75 milliseconds. The time interval between write commands (from loss to acquisition) will be a minimum of 100 milliseconds.

b. Deflection command characteristics - SIDS Azimuth and Elevation Deflection signals will be provided in differential form varying between zero and 7 volts, Each wire of the differential pair ranges from +0.6 to +7.6 volts referenced to aircraft frame ground, The EDSCG shall be capable of processing slew rates up to 400 volts per second. The scale factor of the deflection commands with respect to the armament datum line (which corresponds to display center) shall be 0.28 volt/degree (16 mV/mrad).

c. Steering symbol characteristics - The steering symbol shall be presented on the display as a rectangle 1/160 of the display wide approximately 0.025 inch (0.064 cm) and 1/128 of the display high approximately 0.031 inch (0.079 cm) . It shall be displaced from the center of the display in proportion to the amplitude of the deflection commands.

Upon receipt of the SIDS Available Ground discrete, a SHRIKE advisory light shall be illuminated on the CRTD and the EDSCG shall automatically switch to display SIDS information unless overridden. The SIDS display shall have precedence over all other displays including TV and FLIR. The capability shall be provided to manually override the automatic selection of the SIDS display only via the OVERRIDE switch located on the Forward CRTD; the function of this switch is detailed further in 3.5.2.4.7. A complete matrix of aircraft sensor priorities is specified in Table XI.

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3.4.3.8 TV display - The EDSCG shall accept a composite TV signal consisting of video and horizontal and vertical sync pulses and display the video content on the CRTD. The TV display format shall be 525 line raster scan consisting of two positive 2:1 interlaced fields with a 60 Hz update rate. The raster size shall be a 4 inch by 4 inch (10.16 cm by 10.16 cm) square minimum, but the raster edges shall be at the tangent point to the edge of the usable viewing surface of the CRT +1/16, -0 inch (+0.16, -0 cm). The composite video signal provided will conform to MIL-G-81132. Upon receipt of the TV Available Ground discrete, a TV advisory light shall be illuminated on the CRTD and the EDSCG shall automatically switch to display TV information unless overridden. The TV display shall have precedence over all other displays except SIDS. The capability shall be provided to manually override the automatic TV display selection via the OVERRIDE switch as specified in 3.5.2.4.7 and 3.5.3.4.7. A complete matrix of aircraft sensor priorities is provided in Table XI.

3.4.3.9 FLIR display - The EDSCG shall have the capability of displaying 875 line FLIR information. The 875 line composite signal shall conform to RS-343-A except as modified in this paragraph. The horizontal active time of the display shall conform to the horizontal active time specified in RS-343-A, but 5 to 11 percent less active time will be allowed for sweep retrace. The FLIR composite video shall be transmitted to the EDSCG differentially on two coax lines and shall drive differential line receivers which present a characteristic impedance of 75 ohms line-to-line (37.5 ohms per line). The video amplifier specified in 3.4.11.4 shall be operational whenever FLIR sensor information is being displayed on the CRTD(s). The detail characteristics of the FLIR input shall be as follows:

a. Frame rate - 30 per second

b. Field rate - 60 per second

c. Interlace - 2:1

d. Lines per frame - 875

e. Active lines per frame - 809

f. Aspect ratio - 1:1

Upon receipt of the FLIR Available Ground discrete, the TV advisory light shall be illuminated on the CRTD and the EDSCG shall automatically switch to display FLIR information unless overridden. The FLIR display shall have precedence over all FLR displays, but shall be subordinate to SIDS or TV display. The capability shall be provided to manually override the automatic FLIR display selection via the OVERRIDE switch as specified in 3.5.2.4.7 and 3.5.3.4.7. A complete matrix of aircraft sensor priorities is provided in Table XI.

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3.4.3.10 Standby operation - The CRTD shall be capable of full operation within 60 seconds after application of primary power (3Ø 115 VAC and 28 VDC aircraft power). After 60 seconds and upon receipt of a SIDS Available Ground discrete, a TV Available Ground discrete, or a FLIR Available Ground discrete, the CRTD shall present the commanded display unless the OVERRIDE switch is in OVERRIDE. In the case of multiple sensor availability, normal sensor priorities shall be in effect in STANDBY mode. The indicator OVERRIDE function and sensor priority shall be as specified in Table XI. While in standby, operation in a "RDR" display condition as specified in Table XI shall result in a blanked (darkened) CRT except that fault isolation and test modes requiring display observation shall command an unblanked display while in STANDBY following radar timeout.

3.4.3.11 Display erasure - The SDC memory shall be automatically cleared and the display erased before storing fresh data when either the selected range scale or FLR operating mode is changed, except if HOLD mode is in effect. The SDC memory shall also be cleared when entering or exiting the EXPAND mode and SIDS mode, and when exiting the HOLD mode. Unless HOLD mode is in effect, the SDC memory shall be cleared when exiting the TV and FLIR display modes whenever transmitter silence has been invoked in accordance with Table XI.

3.4.4 Range scale selection - An external switch closure discrete shall be provided to the SDC as a cue to change the range scale selected in all PPI modes. (Closure of this switch shall also be used to reset the range tracker to maximum range when in AGR mode.) Closure of this switch in TF, AGR, HOLD, and EXPAND modes shall have no effect on the displayed range. In EXPAND mode, the range scale light shall change. To prevent crosstalk in the dual indicator TA7C and A-7K EDSCG configurations when the range scale advisory lights are dimmed, separate isolated range advisory lamp excitation signals shall be provided by the SDC for corresponding range scale lights. When EXPAND mode is exited, the display shall return to the scale selected on the range scale lights. In AGR and TF modes, the 10 NM advisory lamp shall be illuminated. In HOLD mode, activation of the range scale selection switch shall not cause a change in the status of any of the range scale advisory lamps. The 5 or 10 NM ranges only shall be selectable in TA, CSTA, and CSGMP modes.

3.4.4.1 Range scale memory and preselection - While operating in AGR, TF, TA, CSTA, or CSGMP, the SDC shall remember the last range scale selected during operation in GMP, GMS, or BCN and shall re-enter that range upon return to GMP, GMS, or BCN operation. Additionally, the SDC shall have limited memory of operating range in TA, CSTA, or CSGMP modes as follows:

a. If GMP, GMS, or BCN mode has not been commanded since the last instance of operation in TA, CSTA, or CSGMP, then entry into TF or AGR shall cause the SDC to remember the last range scale selected in TA, CSTA, or CSGMP, and upon re-entry into TA, CSTA, or CSGMP, operation shall resume in the remembered range.

b. If GMP, GMS, or BCN mode has been commanded since the last instance of operation in TA, CSTA, or CSGMP, then the initial range scale upon entry into TA, CSTA, or CSGMP shall be as follows:

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(1) If GMP, GMS, or BCN mode is exited while operating in 10, 20, 40, or 80 NM range, the initial preselected range for TA, CSTA, or CSGMP shall be 10 NM.

(2) If GMP, GMS, or BCN mode is exited while operating in 5 NM range, the initial preselected range scale for TA, CSTA, or CSGMP shall be 5 NM.

3.4.4.2 FLR operating range scale - When the 5 or 10 NM range or EXPAND mode is selected, the fast scan discrete (28V A,B/GMP, GMS = HI) and high PRF command (28V C,D,E/BCN = LO) shall be issued by the SDC, except in BCN mode. When the 20, 40, or 80 NM range has been selected or the BCN mode discrete is received, the slow scan discrete (28V A,B/GMP, GMS = LO) and medium PRF command (28V C,D,E/BCN = HI) shall be issued by the SDC.

3.4.5 Air-to-ground ranging (AGR) - The EDSCG shall provide the capability to search, acquire and track the range of video targets which are input from the FLR. The AGR function shall track either point (isolated) or distributed (clutter) type targets utilizing the elevation monopulse Sum and Bi-polar video provided by the FLR. Automatic decision making shall be utilized to determine which mode of tracking (point or clutter) is to be used, based upon the video target characteristics as specified in 3.4.5.2 or 3.4.5.3. Provisions shall be made for resetting the range tracker to maximum range upon the external switch closure specified in 3.4.4.

3.4.5.1 AGR processor operation - Upon selection of the AGR mode, the range tracker shall commence a search in range from 60,000 feet (18,288m) to 1,400 feet (427m) and lock-on the first target of sufficient strength. Failure to lock-on prior to reaching 1,400 feet (427m) shall result in the range tracker recycling to 60,000 feet (18,288m) and starting another search cycle. The AGR clutter mode shall utilize the elevation monopulse Sum and Difference video to provide tracking of clutter targets by positioning a sample gate over the crossover point of the Difference video and generating the range output by time measurement to the gate position. Sum channel video shall be used as a threshold function to enable the system to lock-on only when a target return of sufficient strength for tracking is present. The AGR point mode shall utilize Sum video to provide ranging against discrete targets less than 1,080 feet (329m) long in range. An external switch closure shall be provided as a command to the EDSCG to discontinue tracking, reset the range tracker to 60,000 feet (18,288m) and begin a new search cycle. The FLR transmitter power fail/safe discrete shall not affect the operation of the AGR mode.

3.4.5.2 Tracking rate and accuracy (clutter targets) - The EDSCG shall be capable of tracking video targets at range rates of at least 4,000 feet per second (1219m/sec). Tne range accuracy for clutter targets (assuming radar parameters defined in MIL-F-81898) at range rates of 4,000 feet per second (1219m/sec) or less is specified in Table XII.

3.4.5.3 Tracking rate and accuracy (point targets) - For isolated targets, the EDSCG shall be capable of locking-on and tracking sum video signals of 2.8 $\pm$ 0.1 volts, 0.18 microsecond minimum width at tracking rates up to 4,000 feet per second (1219m/sec) independent of depression angle with the following accuracy:

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a. ±25 feet (±7.6m) from 1,000 feet (305m) to 10,000 feet (3050m) (1σ)

b. ±0.25 percent of range to target from 1,000 feet (3050m) to 60,000 feet (18,300m) (1σ)

3.4.5.4 AGR lock-on requirements - The range tracker shall maintain track out to 60,000 feet (18,288m) at grazing angles above 10 (0.17 rad) for clutter targets; otherwise, lock-on shall be maintained at ranges specified in 3.4.5.2 and 3.4.5.3 under the conditions specified.

3.4.5.5 Range set discrete - The 28 VDC range set discrete shall not be provided by the EDSCG. The EDSCG shall provide only the range set return signal.

3.4.6 EDSCG/NWDC data link - Digital interface electronics shall provide the capability for a continuous serial transfer of information between the EDSCG and NWDC. The information content and transmission format shall be as specified in the following paragraphs.

3.4.6.1 Data channel architecture - Information shall be transmitted over a data channel which consists of four twisted pair lines and one signal return line. Each twisted pair line shall consist of a true and a complement signal line. The four twisted pair lines transmit the following signals:

a. Synchronization Clock

b. Data Ready Word

c. Data Word

d. Identification Word

3.4.6.2 Data channel signal characteristics - Signals transmitted over a data channel shall have the following characteristics.

a. Rise time

Maximum 2 microseconds

b. Fall time

Maximum 2 microseconds

c. Pulse coincidence

± 1.0 microsecond between signals; ± 0.1 microsecond between trues and complements of each signal

d. Digital (binary) zero state

+5.0 ± 1.0 volts

e. Digital (binary) one state

0.0 ± 0.5 volts

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3.4.6.3 CDCE data channel - The EDSCG shall receive altitude, cursor azimuth, and cursor range information from the NWDC. The CDCE data channel signals are formatted as follows:

a. Synchronization Clock

Pulses at a frequency of 50 ± 5 kHz.

b. Data Ready Word

The Data Ready Word shall be one bit length in duration and shall occur concurrent with bit number 1 of the Data Word. The leading edge of the Data Ready Word shall be concurrent with the leading edge of the Synchronization Clock Pulse.

c. Data Word

Each Data Word (altitude, azimuth, range) shall be formatted as a 20-bit word. The bit designation shall be as follows:

BIT	DESIGNATION
1	Control bit
2	Validity bit
3-18	Data bits including sign in order of increasing weight. Bit significance for each word type is specified in Table XIII.
19	Unused (digital zero)
20	Parity (odd)

d. Identification Word

The Identification Word shall be transmitted simultaneously with each Data Word and serves to identify the type of information in the Data Word. The bit designation shall be:

BIT	DESIGNATION
1	Control bit (digital one)
2-13	Unused (digital zero)
14-15	Data identification code as specified in Table XIII.
16-19	Unused (digital zero)
20	Parity bit (odd)

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3.4.6.3.1 SDC receiver/detector requirements - A differential input circuit shall be used to receive the true and complement signal pairs from the NWDC. The EDSCG shall check for the presence of the Data Word Validity Bit, for odd parity on the Data Word and Identification Word, and for the proper word identification code and shall process the incoming data if all of these checks are successful.

3.4.6.3.2 Data word order of transmission - The Data Word and ID Word shall be transmitted in the order: Range Word, Azimuth Sine Word, Altitude Word.

3.4.6.4 AGR data channel - The EDSCG shall transmit AGR track range, certain mode information, and display subsystem identification information to the NWDC. The Data words transmitted by the EDSCG shall be synchronized with the Synchronization Clock pulses received from the NWDC over the CDCE Data Channel as specified in 3.4.6.3. In the event the EDSCG receives a NWDC Fail discrete, the Data Word shall be synchronized with an internal back-up clock. The AGR data channel signals are formatted as follows:

a. Synchronization Clock

Pulses at the same frequency as the Synchronization Clock pulses received over the CDCE Data Channel. In the event that the EDSCG receives a NWDC Fail discrete, the frequency shall be 50 $\pm$ 5 kHz generated internally by the EDSCG.

b. Data Ready Word

The Data Ready Word shall be one bit length in duration and shall occur concurrent with bit number 1 of the Data Word. The leading edge of the Data Ready Word shall be concurrent with the leading edge of the Synchronization Clock Pulse.

c. Data Word

The Data Word shall be formatted as a 20-bit word. The bit designation shall be as follows:

BIT	DESIGNATION
1	Control bit (digital one)
2	Validity bit (AGR lock-on or EDSCG identification)
3-5	Unused (digital zero)
6-17	Slant range information. Bit 6 is the LSB (20 feet), Bit 17 is the MSB (40,960 feet)
18	Data sign bit. Positive is a digital zero
19	Unused (digital zero)
20	Parity (odd)

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d. Identification Word

The Identification Word shall be transmitted simultaneously with each Data Word and serves to identify the Data Word, Certain mode information and display subsystem identification information shall also be contained in this word. The bit designation shall be:

BIT	DESIGNATION
1	Control bit (digital one)
2-13	Unused (digital zero)
14	AGR mode or 40 NM, 80 NM identification bit. (Digital one in AGR mode and digital one in 40 NM and 80 NM ranges)
15	BCN identification bit (digital one in BCN mode)
16	SDC ID (digital one constantly)
17	Expand (digital one in Expand only)
18	Hold identification bit (digital one in Hold)
19	Unused (digital zero)
20	Parity bit (odd)

3.4.6.4.1 AGR lock-on/EDSCG I.D. bit - Data Word bit 2 shall indicate AGR lock-on (VALIDITY) status when the FLR is operating in AGR modes (MANUAL or SLAVE). Digital one shall indicate valid radar lock-on and digital zero shall indicate a failure to lock-on. The slant range to the video target being tracked by the AGR processor specified in 3.4.5 shall be contained in bits 6 through 17 of this word. Bit 2 shall be set to a digital one in all other modes, providing the SDC has detected the presence of an "enhanced" Power Supply Programmer and an "enhanced" Forward CRTD via the interface protection circuitry specified in 3.4.11.1. In addition, Data Identification Word bit 16 shall always be set to a digital one. The occurrence of both of these bits in a digital one state shall indicate to the NWDC the presence of an EDSCG.

3.4.6.4.2 AGR/40+80 NM bit - Data Identification Word bit 14 shall indicate either operation in AGR mode or selection of the upper range scales. Presence of a digital one in this bit position shall cue the NWDC of FLR operation either in AGR modes (MANUAL or SLAVE) or in 40 NM or 80 NM range scale.

3.4.6.4.3 BCN bit - Data Identification Word bit 15 shall cue the NWDC of FLR operation in BCN mode (Ground or Airborne) when set to a digital one.

3.4.6.4.4 EXPAND bit - Data Identification Word bit 17 shall cue the NWDC of FLR operation in EXPAND mode when set to a digital one.

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3.4.6.4.5 HOLD bit - Data Identification Word bit 18 shall cue the NWDC of FLR operation in HOLD mode when set to a digital one. This shall indicate FLR operation with a frozen display and silenced transmitter.

3.4.7 Alarm monitor - The EDSCG shall provide the capability to monitor radar and other aircraft system FAIL/SAFE signals and provide indications of unsafe operation in TF, TA, CSTA, and CSGMP modes.

3.4.7.1 Radar TF failure monitoring - The following signals shall be monitored for indication of a failure or unsafe condition when TF, CSTA or CSGMP modes are selected. If any of these signals drop below the Safe level, the EDSCG shall output a 28 volt DC Fail Override signal to the radar and remove the ADI horizontal pointer flag excitation (312 millivolts minimum into a 1000 ohm load). The FAIL light on the CRTD(s) shall also be illuminated.

a. Transmitter power fail/safe

(18 volts DC nominal = Safe, 0-10 volts DC = Fail)

b. Receiver AFC fail/safe

(18 volts DC nominal = Safe, 0-10 volts DC = Fail)

c. Antenna scan fail/safe

(18 volts DC nominal = Safe, 0-10 volts DC = Fail)

d. Receiver-computer fail/safe

(18 volts DC nominal = Safe, 0-10 volts DC = Fail)

e. Test pulse fail/safe

(18 volts DC nominal = Safe, 0-10 volts DC = Fail)

f. Servo bit

(15 volts DC nominal = Safe, 0-10 volts DC = Fail)

3.4.7.2 Radar TA failure monitoring - In TA mode, the EDSCG shall monitor the Transmitter Power Fail/Safe and Receiver AFC Fail/Safe signals described in 3.4.7.1 and the FAIL light on the CRTD(s) shall be illuminated if either signal indicates a FAIL condition.

3.4.7.3 Aircraft system failure monitoring - The signals specified in "a." through "d." following shall be monitored by the EDSCG to determine unsafe flight conditions in TF, CSTA, and CSGMP modes. If any one of these signals drops below the Safe level, the EDSCG shall output a 28 volt Fail Override signal to the radar and remove the ADI horizontal pointer flag excitation. The FAIL light on the CRTD(s) shall not be illuminated due to any of these signals indicating a FAIL. The SDC AGR/TF test switch as described herein, shall have the capability to override the following Fail/Safe signals, except excessive turn rate, to a Safe condition.

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a. Radar altimeter fail/safe

(4 volts minimum = Safe, 1 volt max = Fail)

b. NWDC computer fail/safe

(20 VDC minimum = Safe, open cirucit = Fail)

c. IMS fail/safe

(20 VDC minimum = Safe, open circuit = Fail)

d. Excessive turn rate

(20 VDC minimum = Safe, 1 volt max = Fail)

3.4.8 Fault isolation - The EDSCG shall provide the capability to isolate failures within the FLR to the faulty WRA based upon signal inputs from the FLR. Additional fault location capability shall be included to allow fault isolation within the EDSCG to the faulty WRA as specified in 3.3.7.4. All FLR and EDSCG fault location tests shall be performed with the FLR operating in STANDBY mode. Upon receipt of the 28V in Fault Isolation discrete, the FAIL light on the CRTD(s) shall be illuminated. GO/NO GO operational status check capability of miscellaneous avionics interfaces critical in AGR/TF modes shall be provided via special selftest functions.

3.4.8.1 FLR fault location modes - The EDSCG shall provide monitoring functions of the FLR signals specified in 3.4.8.1.1 through 3.4.8.1.7. Independent from aircraft system inputs (other than from the FLR and aircraft transformers), the EDSCG shall accept mode override discretes from the Fault Locator WRA of the FLR and shall provide FAIL/SAFE outputs as defined for each mode of the Fault Locator switch. Mode override logic and sequencing requirements shall be as defined for each test. Upon receipt of a FAIL/SAFE discrete from the Fault Locator, the EDSCG shall provide a lamp driver output for the Fault Locator SAFE lamp. Current handling capability for this lamp driver shall be adequate to drive an equivalent load of two each MS25237-327 lamps.

3.4.8.1.1 Aircraft transformer test - The EDSCG shall monitor the 11.8 VAC, 26 VAC ØC, 26 VAC ØC/-90° and 28.6 VAC inputs from the aircraft reference transformers for proper levels and provide a logical zero output signal designated "ACFT FAIL/SAFE" if all signals are present.

3.4.8.1.2 Power Supply Programmer Test - Upon receipt of a 28 VDC Mode Override/CSTA discrete in conjunction with a 28 VDC Turn Rate Simulation discrete, the EDSCG shall provide a 28 VDC Mode Override discrete, a 28V CSTA discrete and a 165 $\pm$ 5 mv ØC Turn Rate simulation signal to the FLR. The following signals shall be monitored and a logical zero discrete shall be provided if all signals are present at their proper levels:

a. +20 volt, -20 volt, 26 volt, and +200 volt power supplies

b. Roll, Azimuth and Elevation motor drive (test points)

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c. Repeater Fail/Safe

d. Computer Unblank Pulses

e. Indicator Blanking

f. End of Scan Pulses

g. Pre-Master Trigger (PMT) Pulses

h. Dither Output (triangle wave)

NOTE: Upon receipt of AZ/EL Generator FAIL, the following tests shall be deleted from the power supply test:

(1) Computer Unblank Pulses

(2) Indicator Blanking

(3) End of Scan Pulses

3.4.8.1.3 Transmitter test - Upon the receipt of a 28 Volt Mode Override/CSTA/Load discrete signal, the EDSCG shall output the corresponding mode logic to the FLR and shall monitor the following signals. A logical zero output signal, designated "XMTR FAIL/SAFE" shall be provided to the FLR if all signals are present at their proper levels:

a. Position Analog (tuner feedback)

b. Master Trigger (detected RF pulse)

c. Transmitter Test Function (transmitter power level)

3.4.8.1.4 Computer test pulse test - Upon receipt of a 28 VDC Mode Override/CSTA discrete in conjunction with a 28 VDC Simulation Command, the EDSCG shall provide the corresponding output discretes to the FLR and monitor the Test Pulse Test Function signal. If the signal level is greater than 10 volts, a logical zero discrete, designated "TEST PULSE FAIL/SAFE", shall be provided to the FLR. At the time the 28 VDC Simulation Command is received, the EDSCG shall provide outputs on the radar altitude, ground track velocity, and flight path angle output lines as follows:

a. Radar altitude: 4.0 ± 0.1 VDC

b. Velocity: 2.7 ± 0.1 VDC

c. Climb angle: 0 VDC (Hi and Lo shorted together)

NOTE: The SDC AGR/TF test switch as described herein shall also be capable of providing the signal substitutions in "a.", "b." and "c."

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3.4.8.1.5 Antenna-Receiver test - Upon receipt of a 28 VDC Mode Override/ Antenna Receiver discrete in conjunction with a 28 VDC Simulation Command, the EDSCG shall provide 28 VDC Simulation Command and 28 VDC Mode Override discretes to the FLR and sequence through the following tests:

a. Monitor roll motor drive

If satisfactory, issues 28 VDC CSTA command,

b. Monitor antenna scan, AZ and EL generator and receiver bit

If all signals are satisfactory, issues 28 VDC Load command.

c. Monitor AFC test function

If satisfactory, removes 28 VDC CSTA command and 28 VDC Load command and issues a 28 VDC Beacon mode command.

d. Monitor beacon bit

If satisfactory, a logical zero discrete, designated "A/R FAIL/SAFE," shall be provided to the FLR.

3.4.8.1.6 Computer test no. 2 - Upon receipt of a 28 VDC Mode Override/CSTA discrete in conjunction with a 28 VDC Simulation Command, the EDSCG shall provide the same discrete outputs and simulated aircraft signals as specified in 3.4.8.1.4. In addition to monitoring the Test Pulse Test Function signal, the EDSCG shall monitor the Computer/Receiver Test Function signal and if both signals exceed a 10 volt command, a logical zero discrete, designated "COMPUTER FAIL/SAFE", shall be provided to the FLR.

3.4.8.1.7 Computer test no. 3 - Upon receipt of a 28 VDC Mode Override/CSTA discrete in conjunction with a 28 VDC Climb Angle Simulation Command, the EDSCG shall provide the same discrete outputs as specified in 3.4.8.1.4. The EDSCG shall monitor the same input signals as specified in 3.4.8.1.6 and provide the same COMPUTER FAIL/SAFE discrete to the FLR if both signals are the proper level. At the time the 28 VDC Climb Angle Simulation Command is received, the EDSCG shall provide outputs on the radar altitude, ground track velocity, and flight path angle output lines as follows:

a. Radar altitude: 4.0 $\pm$ 0.1 VDC

b. Velocity: 2.7 $\pm$ 0.1 VDC

c. Climb angle: 10.7 $\pm$ 0.1 VDC

NOTE: The SDC AGR/TF test switch as described herein shall also be capable of providing the signal substitutions in "a.", "b." and "c."

3.4.8.2 EDSCG fault isolation - The EDSCG shall provide outputs to the FLR Fault Iocator to indicate the GO/NO GO status of the EDSCG BIT circuits (power supplies, etc,) and the GO/NO GO status of the DSC processor. The operational status of the CRTD shall be indicated by appropriate displays during fault isolation operations of the FLR and EDSCG.

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3.4.8.2.1 BIT test - The EDSCG shall monitor the operational readiness of the BIT power supplies and other circuits necessary to assure proper BIT operation. A logical zero discrete, designated BIT SAFE, shall be provided at all times when these circuits are operating properly.

3.4.8.2.2 DSC test - Upon receipt of a 28 VDC MODE OVERRIDE & TA discrete, the DSC shall issue to the FLR whatever mode logic is necessary to assure an end-to-end check of the DSC. The existing "SWP GEN" position on the Fault Locator WRA shall be used for this test. As a minimum, an artificial video pattern shall be generated and loaded into the DSC memory. The resulting composite video shall be displayed on the CRTD(s) to aid in isolating failures. In addition to displaying the pattern, the DSC shall monitor the composite video pattern and ensure that, as a minimum, maximum and minimum output video levels correspond to gray shade 7 and gray shade 0 respectively. If all functions check satisfactorily, a logical zero discrete output designated SWEEP GENERATOR SAFE shall be provided. Logical zero is SAFE; logical one is FAIL.

3.4.8.3 AGR/TF self-test - A three position switch shall be provided on the SDC to allow the independent selection of: (1) AGR Test 1 or TF Test 1, (2) OFF (No test-normal operation), and (3) AGR Test 2 or TF Test 2. The switch shall be easily accessible when the SDC is installed in the aircraft and shall be spring-loaded to the OFF position. Actuation of the switch shall not cause a simulation command, a mode override command, a fail override signal, a 28V AGR Test 1 discrete, or a 28V AGR Test 2 discrete to be output by the SDC.

3.4.8.3.1 AGR test no. 1 - This test shall verify the integrity of the AGR Data Channel interface between the EDSCG and the NWDC. The test mode shall be enabled upon receipt of a 28 VDC discrete designated AGR Test No. 1 while operating in AGR mode. During this test mode, the SDC data transmitter shall continuously transmit a digital range word corresponding to a value of 10,000 feet (3048m). The validity bit of the range Data Words transmitted shall be set to simulate a locked condition.

3.4.8.3.2 AGR test no. 2 - This test shall verify basic operation of the SDC AGR processor. The test mode shall be enabled upon receipt of a 28 VDC discrete designated AGR Test No. 2 while operating in AGR mode. During this test mode, the SDC AGR processor shall lock on a simulated video test signal generated by Set Control WRA 10. The apparent range of the test signal shall be selectable via the Range Set thumbwheel switches on WRA 10. Once the SDC has determined the test target slant range, this information shall be transmitted to the NWDC.

3.4.8.3.3 AGR tests via SDC test switch - AGR Test 1 and Test 2 commanded via the AGR/TF TEST switch shall cause initiation of the same functional tests as presently commanded by the externally generated AGR Test 1 and Test 2 signals. AGR self-testing via the AGR/TF TEST switch shall only be available when the EDSCG is operating in the AGR mode. Operation in AGR self-test mode via the AGR/TF Test switch shall continue only as long as the TEST switch is held in the selected test position.

3.4.8.3.4 TF tests via SDC test switch - TF Test 1 shall cause simulation of the same Climb Angle, Radar Altitude, and Velocity to be output to the TF Computer as in output in Computer Test 2 (commanded by the Fault Locator). TF Test 2 shall cause simulation of the same Climb Angle, Radar Altitude, and

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Velocity to be output to the TF Computer as is output in Computer Test 3 (commanded by the Fault Locator). TF self-testing via the AGR/TF TEST switch shall only be available when the SDC is operating in TF mode. Operation in TF self-test mode via the AGR/TF TEST switch shall continue for approximately 17 seconds after the TEST switch is actuated or until the switch is returned to the OFF position, whichever occurs first, During TF Test 1 or Test 2, the SDC shall output the 28V TF Test discrete. A "safe" condition for the IMS FAIL, NWDC FAIL/SAFE, and RADAR ALTIMETER FAIL/SAFE signals shall be simulated by the SDC during TF Test 1 and Test 2 to preclude generation of a FAIL OVERRIDE signal and loss of ADI horizontal pointer flag excitation in the event of a "fail" condition being signaled by the IMS, NWDC, or Radar Altimeter; no signal shall be output by the EDSCG on the IMS FAIL, NWDC FAIL/SAFE, or RADAR ALTIMETER FAIL/SAFE lines.

3.4.8.3.5 TF self-test protection - Protective test monitoring logic shall be incorporated in the EDSCG to detect presence of the 28V TF Test discrete during normal operation in TF mode when TF self-test is not being selected. Upon detection of such a failure, the EDSCG shall remove the ADI horizontal pointer flag excitation, output a 28 V FAIL OVERRIDE signal and illuminate the FAIL light on the CRTD(s).

3.4.9 Radar silence - Provision shall be made to silence the FLR transmitter in modes requiring no update of the FLR sensor information.

3.4.9.1 HOLD mode silence - Whenever HOLD mode is selected on the CRTD and is valid for the operating FLR mode, the SDC shall command that the FLR transmitter cease transmitting. Upon determination by the SDC that transmission is no longer occurring, the display shall be frozen. For purposes of range cursor position updates in HOLD mode requiring operation of the DSC processor, the SDC shall utilize the Pre-Master Trigger (PMT) signal in lieu of the absent Master Trigger (MT) signal to initiate DSC processor cycles.

3.4.9.2 TV/FLIR/SIDS mode silence - When TV, FLIR, or SIDS display modes are available and selected for display on the EDSCG CRTD(s), FLR transmitter silence shall be commanded unless the FLR is operating in the AGR Slave mode or a TF mode. In the dual indicator TA-7C or A-7K EDSCG configuration, selection of external sensor displays on both CRTD's shall be required to command transmitter silence in TF/FLIR/SIDS display modes. Proper SDC moding to provide this function in the single indicator A-7E or A-7D aircraft EDSCG configuration shall occur automatically through use of a single line unique to the wiring of these aircraft. This signal line, designated Aft Sensor Select Ground, shall serve to simulate the corresponding signal from the absent Aft CRTD by occurring concurrent with occurrence of the Forward Sensor Select Ground discrete. The interconnect for the additional simulated Aft Sensor Select Ground discrete is listed in Tables VIII and IX.

3.4.10 Auto-tilt - Whenever the EDSCG is operating in a NWDC - controlled (CDCE operate) mapping or Ground BCN mode, Auto-Tilt may be entered. A switch shall be provided on the CRTD(s) to disable operation in Auto-Tilt. Upon receipt of a CDCE Operate command, the SDC shall enable this switch to function by issuing the Auto-Tilt Available discrete to the CRTD(s). Whenever Auto-Tilt operation is valid and selected via the CRTD switch, a discrete, designated Auto-Tilt Command, shall be issued by the CRTD to the FLR Power Supply Programmer WRA to enable a change in antenna elevation stabilization. Should the Auto-Tilt switch be actuated in the absence of the CDCE Operate Command, manual tilt shall remain enabled in Air BCN, GMP, CSGMP, GMS,

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and manual AGR modes. An Auto-Tilt advisory lamp shall be provided on the CRTD(s). The advisory lamp shall illuminate only upon receipt by the CRTD(s) of a discrete designated Auto-Tilt lamp supplied by the FLR Power Supply Programmer. The advisory lamp shall be dimmed from the cockpit dimming control in conjunction with the range scale advisory lamps; therefore, the AutoTilt Lamp excitation signal shall be isolated within the CRTD(s) to prevent crosstalk in the dual-indicator TA-7C and A-7K EDSCG configurations. The signal interconnect for Auto-Tilt is listed in Tables VIII, IX, and X.

3.4.11 Interface signals - The interface interconnect signal requirements for the EDSCG are specified in Tables VIII, IX, and X. Input/Output/interconnecting wiring for the SDC is listed in Table VIII; for the Forward CRTD in Table IX; for the Aft CRTD in Table X.

3.4.11.1 Interface protection circuitry - The EDSCG and FLR Power Supply Programmer shall utilize the five designated "Protection Interface" signals in Table XIV to enable each "enhanced" WRA (SDC, CRTD(s), Power Supply Programmer) to detect the presence or absence of critical "enhanced" WRA's. Only upon detection of the valid "enhanced" configuration will the protected signal lines listed in Table XV become functional. In the event of the presence of one or more "non-enhanced" WRA's, the protected signal lines in the remaining "enhanced" WRA's shall remain electrically isolated from the functional driver/receiver inside the WRA. Attempted operation of a configuration utilizing an "enhanced" Power Supply Programmer WRA in the absence of "enhanced" SDC, an "enhanced" Forward CRTD, or both shall result in interruption of primary power to the entire FLR system.

3.4.11.2 Blanking pulse - The EDSCG shall provide a blanking pulse for use by external equipment. The blanking pulse characteristics shall be as follows:

a. Amplitude

15 ± 3 volts peak

b. Timing

Delay between leading edge (50 percent point) of PMT input and leading edge (50 percent point) of Blanking Pulse output shall be not greater than 100 nanoseconds. Delay between trailing edge (50 percent) of PMT and trailing edge (50 percent point) of Blanking Pulse shall be not greater than 500 nanoseconds.

c. Rise time

200 nanoseconds maximum (10 percent to 90 percent)

d. Fall time

500 nanoseconds maximum (90 percent to 10 percent)

e. Load impedance

93 ohms (nominal)

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3.4.11.3 Recorder composite video - The EDSCG shall provide a buffered composite video output for driving a video tape recorder in all FLR and SIDS display modes. The characteristic impedance of the output shall be 75 ohms. The field rate of the output video shall be 60 ± 0.1 Hz.

3.4.11.4 Video amplifier enable - Upon receipt of a discrete ground signal, designated Video Amplifier Enable Ground, the SDC shall internally amplify composite video input to it from other sensors external to the FLR system. This amplifier shall have the capability to amplify 525 line an 875 line composite signals from 1 1/4 ± 1/4 volts peak-to-peak to a level compatible with the display requirements of the CRTD(s). The interconnect for the Video Amplifier Enable Ground Signal is listed in Table VIII.

3.5 Detail requirements - Details of all WRA outline and mounting provisions shall be as follows:

3.5.1 Converter, Signal Data CV-3779/APQ-126(V) - The SDC shall meet the following requirements:

3.5.1.1 Function - The SDC shall contain the necessary circuitry for radar video processing, air-to-ground ranging computations and data transmission, symbology generation, and FLR BIT processing. Additional circuitry necessary for interface with miscellaneous aircraft and FLR system functions shall also be located in the SDC.

3.5.1.2 Form factor - The SDC shall be contained within the following dimensions:

a. Height 6,56 inches (16,66 cm)

b. Width 9.53 inches (24.21 cm)

c. Depth 22.08 inches (56,08 cm)

3.5.1.3 Weight - The weight of the SDC shall be not greater than 50 pounds (22.7 Kg).

3.5.1.4 Controls - The SDC shall contain no in-flight controls. A test switch, designated AGR/TF TEST, shall permit selection of two ground test modes, designated TEST 1 and TEST 2, in both AGR and TF modes of operation.

3.5.1.5 Electrical connections - Connections to external circuits shall be provided as specified in Table XVI.

3.5.1.6 Mounting - The SDC shall be hard mounted to the airframe behind the cockpit area.

3.5.2 Indicator, Intratarget Data, Forward IP-1418/APQ-126(V)(red lighting) and IP-1443/APQ-126(V) (white lighting) - The Forward CRTD shall meet the following requirements:

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3.5.2.1 Function - The CRTD shall be capable of displaying FLR video returns, beacon returns, AGR symbology, SIDS information, and both 525-line and 875-line composite video information on a 5-inch (12.7 cm) high-resolution CRT. The viewable display area shall be circular with a diameter no less than 4 inches (10.16 cm). The CRTD shall incorporate a low-glare filter in front of the CRT. Neither etched scales nor a variable light polarizing filter assembly shall be incorporated in the CRTD. The filter shall be compatible with the spectral characteristics of all CRT phosphor types used.

3.5.2.2 Form factor - The Forward CRTD shall be contained within the following dimensions:

a. Height 5.81 inches (14.76 cm)

b. Width 7.05 inches (17.91 cm)

c. Depth 19.55 inches (49.66 cm)

3.5.2.3 Weight - The weight of the Forward CRTD shall be not greater than 30 pounds (13.6 Kg).

3.5.2.4 Controls - The Forward CRTD shall have the following controls.

3.5.2.4.1 Brilliance potentiometer - This control, designated BRIL, shall permit operator adjustment of the brightness of any displayed raster. With the CONTRAST control adjusted fully counterclockwise (off), the raster intensity shall be variable from CRT cutoff to 20 percent of maximum attainable CRT brightness.

3.5.2.4.2 Contrast potentiometer - This control, designated CONTRAST, shall permit operator adjustment of the display video amplitude,

3.5.2.4.3 Receiver gain potentiometer - This control, designated RCVR GAIN, shall permit operator adjustment of the gain of the FLR receiver in beacon and map modes only from minimum to maximum by varying voltage output from 15 to 0 volts respectively. Receiver response shall provide reduced sensitivity in the low gain region with increased sensitivity in the high gain region. The receiver gain control circuit shall be designed for compatible operation with a receiver load of 1.73K ohms. The DC control voltage stimulus provided by the circuit shall vary non-linearly with rotation of the control, The control stimulus shall conform in general to the nominal characteristics as follows:

$$V = \frac{\frac{15}{5x} - \frac{0.6}{1.73}}{\frac{1}{5x} + \frac{6}{5(1-x)} + \frac{1}{1.73}}$$

where V = Receiver gain control voltage (volts)

x = Fractional resistance that is set on receiver gain control as follows:

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The fractional resistance (x) is defined by a series of expressions as follows:

$x = 0.0008$ for $R = 0$ to 0.1
$x = 0.0889R - 0.00889$ for $R = 0.1$ to 0.41
$x = 0.00124e^{7.51R}$ for $R = 0.41$ to 0.83
$x = 3.125R - 1.9625$ for $R = 0.83$ to 0.94
$x = 0.9992$ for $R = 0.94$ to 1.0,

where R is the fractional CW rotation of the receiver gain control. The receiver gain control voltage shall fall within $\pm$ 8 percent of the nominal value computed from the defining equation at the specified receiver gain control angle of 50 percent rotation.

When operating with the dual indicator TA-7C and A-7K configurations, control of the receiver gain in the appropriate modes via this control shall be subject to external selection of the function; the controlling cockpit shall have sole control of the receiver gain in these modes.

3.5.2.4.4 Cursor intensity potentiometer - This control, designated CURSOR, shall permit operator adjustment of the range and azimuth cursor intensity and the range readout intensity. This symbology shall be variable in intensity from gray shade 0 to gray shade 7 independent of other video levels. When operating with the dual indicator TA-7C and A-7K configurations, the Forward CRTD cursor control function shall be independent of that of the AFT CRTD.

3.5.2.4.5 NORMAL/HOLD/EXPAND switch - This control, designated NORM/HOLD/EXPAND, shall permit operator selection of unique display modes. Operation in a FLR mode for which a selected display mode is invalid shall result in normal operation and display. Upon selection of the HOLD position in the appropriate modes, a frozen display and transmitter silence shall occur. Upon selection of the EXPAND position in the appropriate modes, range and azimuth display magnification shall occur. This control shall be constantly enabled by the SDC via the 28 VDC switched signal listed in Table IX when operating in the single indicator A-7D/E configurations. Enabling of this control function shall be subject to external selection when operating in the dual indicator TA-7C and A-7K configurations.

3.5.2.4.6 Auto-tilt switch - This control, designated AUTO TILT, shall permit operator selection of antenna tilt mode. Upon selection of Auto-Tilt in CDCE modes (including Ground BCN), antenna tilt shall be controlled automatically by the NWDC; de-selection of Auto-Tilt in CDCE modes shall result in manual antenna tilt control capability via the ANT TILT control on the FLR Set Control WRA. Position of the AUTO TILT control shall be immaterial for non-CDCE modes. This control shall always be enabled by the SDC via the 28 VDC switched signal listed in Table IX when operating in the single indicator A-7D/E configurations. Enabling of this control function shall be subject to external selection when operating in the dual indicator TA-7C and A-7K configurations.

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3.5.2.4.7 External sensor override switch - This control, designated NORM/OVERRIDE, shall permit operator selection/de-selection of automatically prioritized sensor displays. Operation with the control in the NORM position shall result in automatic display of external sensor information when available; operation with the control in the OVERRIDE position shall result in continuous display of FLR return information, regardless of availability of external sensor information. When operating with the dual indicator TA-7C and A-7K configuration, the Forward CRTD sensor override function shall be independent of that of the Aft CRTD, except that the Forward CRTD control alone shall be capable of overriding SIDS displays for both CRTD's.

3.5.2.5 Lamp indicators - The CRTD shall provide the following indicating functions via lamp illumination.

3.5.2.5.1 FAIL indicator - When appropriately excited, the FAIL indicator luminance shall be 150 foot-lamberts (514 candelas/m^2), minimum, aviation red in accordance with FED-STD-3.

3.5.2.5.2 Nautical mile range indicators 5, 10, 20, 40, 80 - When appropriately excited, each indicator shall luminate at 150 foot-lamberts (514 candelas/m^2), minimum, aviation yellow in accordance with FED-STD-3.

3.5.2.5.3 TV available indicator - When appropriately excited, the TV indicator luminance shall be 150 foot-lamberts (514 candelas/m^2), minimum, aviation yellow in accordance with FED-STD-3.

3.5.2.5.4 SHRIKE available indicator - When appropriately excited, the SHRIKE indicator luminance shall be 150 foot-lamberts (514 candelas/m^2), minimum, aviation yellow in accordance with FED-STD-3.

3.5.2.5.5 HOLD mode indicator - When appropriately excited, the HOLD indicator luminance shall be 150 foot-lamberts (514 candelas/m^2), minimum, aviation yellow in accordance with FED-STD-3.

3.5.2.5.6 AUTO TILT mode indicator - When appropriately excited, the AUTO TILT indicator luminance shall be 150 foot-lamberts (514 candelas/m^2), minimum, aviation yellow in accordance with FED-STD-3.

3.5.2.6 Electrical connections - Connectons to external circuits shall be provided as follows:

Reference Designator	Receptacle Type	Function
8J1	Cannon DPKB-71C15-PC3	Power and signals

3.5.2.7 Mounting - The CRTD shall be hard mounted to the airframe in the cockpit.

3.5.3 Indicator, Intratarget Data, Aft IP-( )/APQ-126(V)(red lighting) and IP-( )/APQ-126(V) (white lighting) - The Aft CRTD shall meet the following requirements:

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3.5.3.1 Function - The CRTD shall be capable of displaying FLR video returns, beacon returns, AGR symbology, SIDS information, and both 525-line and 875-line composite video information on a 5-inch (12.7 cm) high-resolution CRT. The viewable display area shall be circular with a diameter no less than 4 inches (10.16 cm). The CRTD shall incorporate a low-glare filter in front of the CRT. Neither etched scales nor a variable light polarizing filter assembly shall be incorporated in the CRTD. The filter shall be compatible with the spectral characteristics of all CRT phosphor types used.

3.5.3.2 Form factor - The Aft CRTD shall be contained within the following dimensions:

a. Height 8.32 inches (21.13 cm)

b. Width 7.06 inches (17.93 cm)

c. Depth 16.42 inches (41.71 cm)

3.5.3.3 Weight - The weight of the Aft CRTD shall be not greater than 30 pounds (13.6 Kg).

3.5.3.4 Controls - The Aft CRTD shall have the following controls.

3.5.3.4.1 Brilliance potentiometer - This control, designated BRIL, shall permit operator adjustment of the brightness of any displayed raster. With the CONTRAST control adjusted fully counterclockwise (off), the raster intensity shall be variable from CRT cutoff to 20 percent of maximum attainable CRT brightness.

3.5.3.4.2 Contrast potentiometer - This control, designated CONTRAST, shall permit operator adjustment of the display video amplitude.

3.5.3.4.3 Receiver gain potentiometer - This control, designated RCVR GAIN, shall permit operator adjustment of the gain of the FLR receiver in beacon and map modes only from minimum to maximum by varying voltage output from 15 to 0 volts respectively. Receiver response shall provide reduced sensitivity in the low gain region with increased sensitivity in the high gain region. The receiver gain control circuit shall be designed for compatible operation with a receiver load of 1.73K ohms. The DC control voltage stimulus provided by the circuit shall vary non-linearly with rotation of the control. The control stimulus shall conform in general to the nominal characteristics as follows:

$$V = \frac{\frac{15}{5x} - \frac{0.6}{1.73}}{\frac{1}{5x} + \frac{6}{5(1-x)} + \frac{1}{1.73}}$$

where V = Receiver gain control voltage (volts)

x = Fractional resistance that is set on receiver gain control as follows:

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The fractional resistance (x) is defined by a series of expressions as follows:

$x = 0.0008$ for $R = 0$ to 0.1
$x = 0.0889R - 0.00889$ for $R = 0.1$ to 0.41
$x = 0.00124e^{7.51R}$ for $R = 0.41$ to 0.83
$x = 3.125R - 1.9625$ for $R = 0.83$ to 0.94
$x = 0.9992$ for $R = 0.94$ to 1.0,

where R is the fractional CW rotation of the receiver gain control. The receiver gain control voltage shall fall within + 8 percent of the nominal value computed from the defining equation at the specified receiver gain control angle of 50 percent rotation.

Control of the receiver gain in the appropriate modes via this control shall be subject to external selection of the function; the controlling cockpit shall have sole control of the receiver gain in these modes.

3.5.3.4.4 Cursor intensity potentiometer - This control, designated CURSOR, shall permit operator adjustment of the range and azimuth cursor intensity and the range readout intensity. This symbology shall be variable in intensity from gray shade 0 to gray shade 7 independent of other video levels. The Aft CRTD cursor control function shall be independent of that of the Forward CRTD.

3.5.3.4.5 NORMAL/HOLD/EXPAND switch - This control, designated NORM/HOLD/EXPAND, shall permit operator selection of unique display modes. Operation in a FLR mode for which a selected display mode is invalid shall result in normal operation and display. Upon selection of the HOLD position in the appropriate modes, a frozen display and transmitter silence shall occur. Upon selection of the EXPAND position in the appropriate modes, range and azimuth display magnification shall occur. Enabling of this control function shall be subject to external selection.

3.5.3.4.6 AUTO-TILT switch - This control, designated AUTO TILT, shall permit operator selection of antenna tilt mode. Upon selection of AUTO-TILT in CDCE modes (including Ground BCN), antenna tilt shall be controlled automatically by the NWDC; de-selection of Auto-Tilt in CDCE modes shall result in manual antenna tilt control capability via the ANT TILT control on the FLR Set Control WRA. Position of the AUTO TILT control shall be immaterial for non-CDCE modes. Enabling of this control function shall be subject to external selection.

3.5.3.4.7 External sensor override switch - This control, designated NORM/OVERRIDE, shall permit operator selection/de-selection of automatically prioritized sensor displays. Operation with the control in the NORM position shall result in automatic display of external sensor information provided in composite video format when available; operation with the control in the OVERRIDE position shall result in continuous display of FLR return or SIDS information, regardless of availability of other external sensor information. The Aft CRTD sensor override function shall be independent of that of the Forward CRTD.

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3.5.3.5 Lamp indicators - The CRTD shall provide the following indicating functions via lamp illumination.

3.5.3.5.1 FAIL indicator - When appropriately excited, the FAIL indicator luminance shall be 150 foot-lamberts (514 candelas/m^2) minimum, aviation red in accordance with FED-STD-3.

3.5.3.5.2 Nautical mile range indicators 5, 10, 20, 40, 80 - When appropriately excited, each indicator shall luminate at 150 foot-lamberts (514 candelas/m^2), minimum, aviation yellow in accordance with FED-STD-3.

3.5.3.5.3 TV available indicator - When appropriately excited, the TV indicator luminance shall be 150 foot-lamberts (514 candelas/m), minimum, aviation yellow in accordance with FED-STD-3.

3.5.3.5.4 SHRIKE available indicator - When appropriately excited, the SHRIKE indicator luminance shall be 150 foot-lamberts (514 candelas/m^2), minimum, aviation yellow in accordance with FED-STD-3.

3.5.3.5.5 HOLD mode indicator - When appropriately excited, the HOLD indicator luminance shall be 150 foot-lamberts (514 candelas/m^2), minimum, aviation yellow in accordance with FED-STD-3.

3.5.3.5.6 AUTO TILT mode indicator - When appropriately excited, the AUTO TILT indicator luminance shall be 150 foot-lamberts (514 candelas/m^2), minimum, aviation yellow in accordance with FED-STD-3.

3.5.3.6 Electrical connections - Connections to external circuits shall be provided as follows:

Reference Designator	Receptacle Type	Function
18J1	Cannon DPKB-71C15-PC3	Power and signals

3.5.3.7 Mounting - The CRTD shall be hard mounted to the airframe in the cockpit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection and test requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections and tests set forth in the specification where such inspections and tests are deemed necessary to assure supplies and services conform to prescribed requirements,

4.2 Classification of inspections - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

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a. First article inspection (See 4.3)

b. Acceptance tests (See 4.4)

4.3 First article inspections - First article shall be conducted on samples of production equipment to determine conformance of the equipment with the requirements of this specification. The equipment samples shall be subjected to all tests of the procuring activity approved first article program, except as otherwise specified in 4.5. The equipment shall successfully pass all the tests to demonstrate equipment conformance to specification requirements,

4.3.1 Sampling - A minimum of two EDSCG subsystems (SDC with mount and CRTD(s)), representative in every way of production equipment, shall be utilized for first article testing. The interconnecting test harness shall be composed of the same types and approximately the same lengths of wire as is used in the aircraft unless otherwise specified in the test method.

4.3.2 Scope of tests - First article tests shall be devised and conducted by the contractor or an approved testing agency. Tests shall include all testing deemed necessary to determine that the equipment satisfactorily meets all the requirements of this specification. First article tests shall include successful completion of environmental tests in accordance with MIL-T-5422 to prove compliance with the requirements specified in 3.3.10 and successful completion of electromagnetic interference tests in accordance with MIL-STD-462.

4.3.2.1 Temperature - altitude test - The equipment shall be tested in accordance with MIL-T-5422 to the applicable performance level of MIL-E-5400 as specified in "a." and "b." following. An acceptance test of the EDSCG system shall be successfully completed prior to the exposure period of this test and shall be successfully completed again after the exposure period. A performance test shall be performed after installation of the EDSCG system into the test facility and at each of the test levels required by MIL-T-5422. In the event of a failure (see 6.5.3), the source of the problem shall be corrected and the testing resumed at the start of the last step successfully completed. Low, nominal, and high primary power shall be 108, 115, and 118 VAC, respectively, for three-phase power and 21, 28, and 30 VDC, respectively for the 28 VDC power.

a. SDC (with mount):

The SDC shall be tested, while installed in its mount, over the range of -54°C to +71°C from sea level to 50,000 feet (15,240 m) as bounded by MIL-E-5400 Figure 6, Sheet 2, Curve A for continuous operation for Class 2 equipment.

b. CRTD(s):

The CRTD shall be tested over the range of -54°C to +71°C from sea level to 50,000 feet (15,240 m) as bounded by MIL-E-5400 Figure 6, Sheet 1, Curves A and B for continuous and intermittent operation, respectively, for Class I equipment.

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4.3.2.2 Electromagnetic interference (EMI) test - The equipment shall be tested in accordance with MIL-STD-462 and shall meet the requirements of MIL-STD461 Notice 3 for Class A1 equipment.

4.3.2.3 Humidity test - The equipment shall be tested in accordance with MIL-T-5422.

4.3.2.4 Vibration test - The equipment shall be tested in accordance with the procuring activity approved contractor test procedure to the applicable performance level outlined in "a." and "b." following.

a. SDC (with mount):

The SDC shall be tested, while installed in its mount, over the ranges of stimuli listed in Table XVII and bounded by the figure indicated.

b. CRTD(s):

The CRTD shall be tested over the ranges of stimuli listed in Table XVIII and bounded by the figures indicated.

4.3.2.5 Shock test - The equipment shall be tested in accordance with MIL-T-5422, except that the levels specified in 3.3.10.5 shall apply.

4.3.2.6 Sand and dust test - The equipment shall be tested in accordance with MIL-T-5422.

4.3.2.7 Fungus resistance test - The equipment shall be tested in accordance with MIL-T-5422.

4.3.2.8 Salt atmosphere test - The equipment shall be tested in accordance with MIL-T-5422.

4.3.2.9 Explosive conditions test - The equipment shall be tested in accordance with MIL-T-5422.

4.3.3 First article test failures - If the sample production EDSCG or WRA fails to pass any test requirement, that particular test shall be halted and the necessary corrective action taken. Full particulars of the failure and the corrective action taken shall be made available to the procuring activity for approval at the time the action is taken by the contractor. Testing may be resumed as soon as the corrective action has been effected. The retesting necessary to demonstrate the environmental suitability of corrective fixes shall be mutually agreed upon by the procuring activity and the contractor. Production drawings, schematics, etc., shall be amended to reflect all agreed upon changes upon completion of first article testing.

4.4 Acceptance tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance test. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activiy. The procuring activity reserves the right to develop and conduct acceptance tests on the production equipment (See 6.2.1). Acceptance or approval of material during the course of manufacture shall not be construed as guarantee of the acceptance of the finished product.

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Acceptance tests shall consist of the following:

a. Individual Tests (See 4.4.1)

b. Reliability Assurance Tests (See 4.4.2)

4.4.1 Individual tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability.

4.4.1.1 Scope of individual tests - Individual tests of EDSCG systems and spare WRA's shall include both environmental burn-in testing and final acceptance testing under room ambient environmental conditions. The procuring activity reserves the right to require modification of the contractor's test program if it does not adequately demonstrate compliance with the procurement specification.

4.4.1.1.1 Environmental burn-in tests - Each EDSCG system and spare WRA shall successfully complete a burn-in testing sequence consisting of environmental and operational cycles as depicted in Figure 11.

4.4.1.1.2 Final acceptance tests - Each item presented for final acceptance testing shall have successfully completed environmental burn-in testing. Each EDSCG system and spare WRA shall successfully complete the final acceptance test without failure (see 6.4.3) or adjustment. Any tests developed and conducted by the procuring activity on the production equipment shall not except the requirements of this specification.

4.4.1.2 FLR integration verification - The first EDSCG subsystem submitted for acceptance under contract shall be tested as part of an assembled "enhanced" AN/APQ-126(V) FLR to demonstrate proper operation in a system environment.

4.4.2 Reliability assurance tests - Reliability Assurance testing shall be in accordance with the contractor's test plan as approved by the procuring activity. (See 6.2.2)

4.5 Qualification by similarity - In lieu of performing any of the testing specified in 4.3.2.1 through 4.3.2.9 and 4.4.2, qualification by similarity comparison shall be acceptable where the similarity claim may be supported in technical detail.

4.6 Test procedures - The procedures used for conducting qualification tests and acceptance tests shall be prepared by the contractor and approved by the procuring activity. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. (See 6.2.1) Individual test procedures shall also be prepared for each subassembly, SRA, or other item designated by the procuring activity as a spare item. (See 6.2.2)

5. PACKAGING

5.1 Packaging and packing - The equipment shall be prepared for delivery in accordance with MIL-STD-794; unit packaging shall be level C and exterior packaging shall be level C, unless specified otherwise in the contract (See 6.2.1).

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5.2 Marking for shipment - Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and shall include the following information.

a. Part name

b. Contractor's name and trademark

c. Contractor's address

d. Month and year of manufacture

e. Date of shipment

6. NOTES

6.1 Intended use - The EDSCG is intended for use as the display subsystem of an "enhanced" AN/APQ-126(V) FLR system. In addition to display of the FLR sensor information, it also provides display symbology generation, AGR processing, FLR fault isolation, alarm monitoring, and miscellaneous other functions. Further, it is intended to serve as an auxiliary display for external sensor information of various formats.

6.2 Ordering data

6.2.1 Acquisition requirements - Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Selection of applicable levels of packaging and packing (see 5.1)

c. If Government will develop and perform acceptance test on production equipment (see 4.4).

d. Test procedures if other than those specified in 4.6.

e. First article (see 3.1.1).

6.2.2 Data requirements - When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Delineated data required by this specification are cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID No.	Option
4.3.3	Failure Analysis Report	DI-R-7041	--
4.4.2	Reliability Test Plan	DI-R-7079	--
4.6	Test Procedures	UDI-T-21347	--

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Associated equipment - Functional characteristics of the equipment listed in Table IV are compatible with complementary functional characteristics of the EDSCG.

6.4 First article - When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

6.5 Definition of terms - The following definitions of terms apply to the interpretation of this specification.

6.5.1 Scheduled maintenance - Scheduled maintenance consists of all inspection, checkout, repair, replacement, or adjustment actions that are performed because of expiration of time, flight hours, cycles or operations, or some special occurrence. These include preflights, postflights, daily, periodic, calendar, phased and special inspections, periodic calibration or adjustments.

6.5.2 Unscheduled maintenance - Unscheduled maintenance consists of all corrective maintenance resulting from observed system/component malfunction or from defects discovered during scheduled inspections. Unscheduled maintenance can be accomplished at three levels of maintenence: Organization, Intermediate, and Depot.

6.5.2.1 Organizational maintenance - Organizational (on aircraft) maintenance consists of verifying and isolating a fault, removing and installing a like serviceable item, making "on aircraft" repairs and performing operational checks to verify proper operation.

6.5.2.2 Intermediate maintenance - Intermediate (off aircraft) maintenance consists of bench check, disassembly, and repair of unserviceable items by replacement of discrepant parts. Intermediate capability is limited only by test equipment, facilities, manpower and training. Serviceability of all items will be verified prior to return to serviceable stocks.

6.5.2.3 Depot maintenance - Depot maintenance consists of repair of system components which exceed the Intermediate level capability for repair and any overhaul.

6.5.3 Failure - For the purpose of this specification, failure is defined as the inability of the equipment to meet all of the operating parameters contained in the procurement specification except for memory failures defined in "a", "b", and "c". Any operating discrepancy that requires an unscheduled adjustment or calibration to be made (except normal operating adjustment or scheduled maintenance procedure), after initial satisfactory operation of the affected equipment, shall be defined as an equipment failure.

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a. Total number of displayed bit failures exceeds 15.

or

b. Two (2) bit failures in adjacent display locations,

or

c. Five (5) bit failures within a 0.5 inch diameter circle on the display.

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TABLE I. WRA interface connectors. 1/

WRA	CONNECTOR REFERENCE	MIL PART NO.	OTHER PART NO.
SDC	6P1	MS27473E24B61S	
	6P2	MS27473E24B61SA	
	6P3		CVC6062E-61-21S-1 2/
	6P4		CVC6062E-61-21SW-1 2/
	6P5	MS27473E24B61SB	
	6P6	MS27473E24B61SC	
Forward CRTD	8P1		202-26222-6 2/
AFT CRTD	18P1		202-26222-6 2/

1/ Connectors listed are those of the aircraft wire harness with which the EDSCG WRA's must mate.

2/ Vought Corporation part numbers.

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Table II. Physically interchangeable units.

EDSCG Units	Previous DSCG Units	Original Units
Converter, Signal Data CV-3779/APQ-126(V)	Converter, Signal Data CV-3368/APQ-126(V)	Sweep Generator SG-811/APQ-126(V)
Indicator, Intra-Target Data IP-1418/APQ-126(V) or IP-1443/APQ-126(V)	Indicator, Intra-Target Data IP-1256/APQ-126(V) or IP-1371/APQ-126(V)	Indicator,Multiple Air Navigation IP-940/APQ-126(V) or IP-952/APQ-126(V)
Indicator, Intra-Target Data IP-( )/APQ-126(V) or IP-( )/APQ-126(V)	Indicator, Intra-Target Data IP-1255/APQ-126(V) or IP-1370/APQ-126(V)	------

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TABLE III. Identification resistor (IDR) values.

REFERENCE DESIGNATOR	IDR 1 VALUE	IDR 2 VALUE
Subassemblies:		
6A1	24.9K	1.62K
6A2	24.9K	26.1K
6A4	24.9K	1.00K
6A6	2.49K	68.1K
6A8	2.49K	1.27K
6A9	2.49K	1.00K
6A10	2.49K	162K
6A11	2.49K	5.90K
6A12	2.49K	7.50
6A13	24.9K	9.53K
6A14	2.49K	71.5K
6A15	2.49K	14.7K
6A16	2.49K	15.4K
6A17	2.49K	75.0K
6A18	2.49K	18.7K
6A19	2.49K	178K
6A20	2.49K	23.7K
6A21	2.49K	78.7K
6A22	2.49K	100K
6A23	2.49K	2.26K
6A24	24.9K	3.01K
6A25	2.49K	105K
6A26	2.49K	110K
6A27	2.49K	187K
6A28	2.49K	147K
6A29	2.49K	154K
Units:		
WRA-6 (6A30) 1/	24.9K	1.07K
WRA-8 (8W1) 1/	4.12K	4.99K
WRA-18 (18W1) 1/	5.90K	4.99K

1/ WRA IDRs are a part of the lower level subassembly identified in parenthesis.

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TABLE IV. Associated equipment.

DESCRIPTION	SPECIFICATION NUMBER	PART NUMBER
FLR WRA's		
Power Supply - Programmer	MIL-F-81898	PP-7799/APQ-126(V)
Transmitter	MIL-F-81898	T-1091/APQ-126(V)
Antenna-Receiver	MIL-F-81898	AS-4072/APQ-126(V)
Air Navigation Computer	MIL-F-81898	CP-954/APQ-126(V)
Mount, Electrical Equipment	MIL-F-81898	MT-4043/APQ-126(V)
Control, Radar Set (WRA 5)	MIL-F-81898	C-11105/APQ-126(V) -or- C-10784/APQ-126(V)
Fault Locator	MIL-F-81898	MX-8195/APQ-126(V)
Control, Radar Set (WRA 10)	MIL-F-81898	C-8253/APQ-126(V) -or- C-8254/APQ-126(V) -or- C-8255/APQ-126(V)
Fan and Duct Assembly	MIL-F-81898	HD-821/APQ-126(V)
Miscellaneous Aircraft Equipment:		
Altimeter, Radar	MIL-A-23887	AN/APN-141(V)
Inertial Measurement Set	VAD 204-16-70 1/	AN/ASN-90(V)
Indicator, Attitude Director	MIL-I-23524	ID-1329/A
Guided Weapon MK39 MOD 0	MIL-G-81132	63A68F1999
Three-Axis Rate Gyro Package	VAD 204-15-24 1/	215-37128 2/
Navigation/Weapon Delivery Digital Computer	MIL-N-85016	AN/ASN-91(V) or AN/ASN-91A
Computer, Air Data	VAD 204-15-26 1/	CP-828/A
Convertor, Signal Conditioner	AS-1720/XWS-6517	CV-2529A/APG
Head-Up Display	VAD 204-16-19 1/	AN/AVQ-7
Transformer	MIL-T-27	CVC 6070-5 2/
Transformer	MIL-T-27	CVC 6070-6 2/
Switch	MS-25089-3	
Switch	MS-25089-3g	

1/ Vought Corporation Specification No.

2/ Vought Corporation Part No.

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TABIE V. SDC cooling air supply characteristics.

ALTITUDE		FLOW RATE (#/MIN)	SUPPLY TEMP. - DB/DAR 1/	
(FT)	(M)		(^{o}F)	(^{o}C)
SEA LEVEL	SEA LEVEL	0.95	45/32	7.2/0
5000	1524	0.88	42/32	5.6/0
10000	3048	0.94	39/32	3.9/0
15000	4572	1.00	36/32	2.2/0
20000	6096	1.04	32/32	0/0
25000	7620	1.04	32/32	0/0
30000	9144	1.00	30	-1.1
35000	10668	0.87	0	-17.8
40000	12192	0.82	-14	-25.6
45000	13716	0.77	-13	-25.0
50000	15240	0.73	-12	-24.4

1/ DB/DAR = Dry Bulb/Dry Air Rated

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TABLE VI. Display mode.

RADAR POWER SWITCH	AIRCRAFT MODE DISCRETE	RADAR MODE SWITCH	EDSCG MODE	CRTD DISPLAY	RANGE SCALES	DISPLAY SCALING	DISPLAY GRAY SHADES	CURSORS	RANGE R/O 1/	AVAILABLE FUNCTIONS	
										HOLD	EXP
OFF	-	-	OFF	BLANK	-	-	-	-	-	-	-
STBY	-	-	STDY 2/	BLANK 2/	-	-	-	-	-	-	-
	OFF	BCN	AIR BCN	±45, PPI	ALL	SLANT RNG	8	MAN RNG	NO	NO	NO
		TA	TA	±45, PPI	5,10	SLANT RNG	8	NO	NO	NO	NO
		GMP	GMP	±45, PPI	ALL	GND RNG	8	MAN RNG	NO	YES	NO
		GMS	GMS	±45, PPI	ALL	GND RING	8	MAN RNG	NO	YES	NO
		AGR	AGR	STROBE	10	SLANT RNG	-	NO	YES	NO	NO
	TFO	BCN	TF	E^2,	10	LOG	2	NO	NO	NO	NO
		TA	CSTA	±20, PPI	5,10	SLANT RNG	8	NO	NO	NO	NO
		GMP	CSGMP	±20, PPI	5,10	GND RNG	8	MAN RNG	NO	YES	NO
		GMS	TF	E^2	10	LOG	2	NO	NO	NO	NO
		AGR	TF	E^2	10	LOG	2	NO	NO	NO	NO
OPR	SLAVE	BCN	AGR	STROBE	10	SLANT RNG	-	NO	YES	NO	NO
		TA	AGR	STROBE	10	SLANT RNG	-	NO	YES	NO	NO
		GMP	AGR	STROBE	10	SLANT RNG	-	NO	YES	NO	NO
		GMS	AGR	STROBE	10	SLANT RNG	-	NO	YES	NO	NO
		AGR	AGR	STROBE	10	SLANT RNG	-	NO	YES	NO	NO
	CDCE	BCN	GND BCN	±45, PPI	ALL	GND RING	8	DRIVEN	YES	YES	YES
		TA	GMP	±45, PPI	ALL	GND RING	8	DRIVEN	YES	YES	YES
		GMP	GMP	±45, PPI	ALL	GND RING	8	DRIVEN	YES	YES	YES
		GMS	GMS	±45, PPI	ALL	GND RING	8	DRIVEN	YES	YES	YES
		AGR	AGR	±45, PPI	ALL	GND RING	8	DRIVEN	YES	YES	YES

1/ Rng R/O = Range Readout

2 Display is blanked while in STANDBY, except while operating in certain BIT Modes which require displays.

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TABLE VII. CRT spot defect acceptance criteria.

TYPE OF SPOT	MAXIMUM DIAMETER (INCH)	MAXIMUM DIAMETER (CM)	NUMBER IN ONE CRT	OF WHICH NOT MORE THAN	ARE MORE THAN (INCH)	ARE MORE THAN (CM)	MINIMUM SEPARATION (INCH)	MINIMUM SEPARATION (CM)
Dead	0.050	0.127	8	4	0.025	0.64	0.250	0.635
Bright	0.040	0.120	4	2	0.025	0.064	0.250	0.635
Color	0.050	0.127	6	2	0.040	0.102	0.025	0.635

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TABLE VIII. SDC interface.

Pin Number	Signal Nomenclature	Signal Interconnect 1/
6P1-A	AUTO-TILT AVAILABLE GND	XA 2/
6P1-C	+20VDC	I
6P1-D	RADAR RETURN	I
6P1-E	-20VDC	I
6P1-F	RADAR SILENCE GND	O
6P1-G	IMS FAIL GND	O
6P1-H	FLIR AVAILABLE GND	XF
6P1-J	DSC GND	XF
6P1-K	28V IN TF	I
6P1-L	28V IN TA	I
6P1-M	28V IN CSTA (IN)	I
6P1-N	28V IN CSGMP	I
6P1-P	28V IN GMP/GMS	I
6P1-R	28V IN GMS	I
6P1-S	28V IN BCN	I
6P1-U	5V NMI BAR	I
6P1-V	28V IN PPI	XA 2/
6P1-X	28V IN AGR	I
6P1-Y	SIDS SELECT GND	XA 2/
6P1-Z	CHASSIS GND	I
6P1-b	28 V IN AB (GMP + GMS)	O
6P1-c	28V IN CDE/BCN	O
6P1-d	28VDC	3/
6P1-e	+5VDC	O
6P1-f	+5VDC RETURN	O
6P1-g	RECEIVER BIT	I
6P1-h	28V IN PPI	XF
6P1-i	NWDC FAIL/SAFE HI	I
6P1-j	INDICATOR BLANKING	I
6P1-k	SHIELD GROUND	I
6P1-m	28V LOAD	O
6P1-n	MODIFIED PSP COMMAND	I
6P1-p	115VAC PH A	I
6P1-q	115VAC PH B	I
6P1-r	115VAC PH C	I
6P1-s	AC GND	I
6P1-t	28V BCN	O
6P1-u	RECEIVER BLANKING	O
6P1-v	RECEIVER BLANKING RETURN	O
6P1-w	ANTENNA AZIMUTH COSINE HI	I
6P1-x	ANTENNA AZIMUTH COSINE LO	I
6P1-y	ANTENNA AZIMUTH SINE HI	I
6P1-z	ANTENNA AZIMUTH SINE LO	I
6P1-AA	ANTENNA AZIMUTH POSITION SHIELD	I
6P1-BB	28V HOLD LAMP	XF
6P1-DD	SIDS SELECT GND	XF
6P1-EE	28 V RNG A	XF
6P1-FF	28V RNG B	XF
6P1-GG	28V RNG C	XF

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TABLE VIII. SDC interface. (Continued)

Pin Number	Signal Nomenclature	Signal Interconnect 1/
6P1-HH	28V RNG D	XF
6P1-JJ	28V RNG E	XF
6P1-KK	26VAC PHC	O
6P1-LL	AC GND	O
6P1-MM	MODE ERASE	I
6P1-NN	CURSOR BRILL	XF
6P1-PP	CURSOR BRILL	XA 2/
6P2-A	SCAN ANALOG (DEMOND)	I
6P2-B	SCAN ANALOG LO	I
6P2-C	28VDC	I
6P2-D	28VDC GND	I
6P2-E	28V GMS/LOAD	O
6P2-F	28V MANUAL LOAD SELECT	I
6P2-G	28VDC	XA 2/
6P2-H	200VDC GND	XA 2/
6P2-J	200VDC	XF
6P2-K	200VDC GND	XF
6P2-L	SENSOR SELECT GND (FWD)	XF
6P2-M	28V LOAD	O
6P2-N	AUTO-TILT AVAILABLE GND	XF
6P2-P	(IDA)	4/
6P2-R	SENSOR SELECT GND (AFT)	XA 2/
6P2-S	115VAC PH A FILTERED	XA 2/
6P2-T	115VAC PH A FILTERED	XF
6P2-U	115VAC PH B FILTERED	XF
6P2-V	115VAC PH C FILTERED	XF
6P2-W	AC GND FILTERED	XF
6P2-X	(IDC)	4/
6P2-Y	28VDC	XF
6P2-Z	28VDC GND	XF
6P2-a	28V IN TA	O
6P2-b	28V IN CSTA (OUT)	O
6P2-c	28V IN CSGMP	O
6P2-d	28V IN GMP/GMS	O
6P2-e	28V IN GMS	O
6P2-g	AC GND FILTERED	XA 2/
6P2-h	115VAC PH B FILTERED	XA 2/
6P2-i	28V IN AGR	O
6P2-j	28V IN TF + MEMORY TEST	XA 2/
6P2-k	28V IN TF + MEMORY TEST	XF
6P2-m	28V NORMAL	XF
6P2-n	TV AVAILABLE GND	XF
6P2-p	28V SIMULATION COMMAND (IN)	I
6P2-q	28V CLIMB ANGLE SIMULATION	I
6P2-r	28VDC GND	O
6P2-s	LAMP DRIVER	O
6P2-t	28V IN FAULT ISOLATION	I
6P2-u	28VDC	O

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TABLE VIII. SDC interface. (Continued)

Pin Number	Signal Nomenclature	Signal Interconnect 1/
6P2-v	28V NORMAL	XA 2/
6P2-w	26VAC PH C/-90 DEG HI	I
6P2-x	AC GND	I
6P2-y	28.6VAC PH C HI	I
6P2-AA	115VAC PH C FILTERED	XA 2/
6P2-BB	DSC GND	XA 2/
6P2-CC	28V HOLD LAMP	XA 2/
6P2-DD	(IDB)	4/
6P2-EE	28V RNG A	XA 2/
6P2-FF	28V RNG B	XA 2/
6P2-GG	28V RNG C	XA 2/
6P2-HH	28V RNG D	XA 2/
6P2-JJ	28V RNG E	XA 2/
6P2-KK	28V FAIL	XA 2/
6P2-LL	28V EXPAND AVAILABLE	XA 2/
6P2-MM	SENSOR SELECT GND (AFT)	XF 5/
6P2-NN	28V FAIL	XF
6P2-PP	28V EXPAND AVAILABLE	XF
6P3-1	FLIR VIDEO HI	I
6P3-2	FLIR VIDEO LO	I
6P3-3	RECORDER VIDEO	0
6P3-4	BLANDING PULSE	0
6P3-5	PMT	I
6P3-6	BIPOLAR VIDEO	I
6P3-7	SUM VIDEO	I
6P3-8	MASTER TRIGGER	I
6P3-9	AFT SELECT VIDEO	0
6P3-10	TV VIDEO	I
6P3-11	TEMPLATE VIDEO	XF
6P3-12	SELECT RADAR VIDEO	XF
6P4-1	FWD SELECT VIDEO	XF
6P4-4	MODIFIED SDC/IND INTERLOCK 6/	XF
6P4-7	MODIFIED SDC GND 6/	XF
6P4-8	MODIFIED SCD/IND INTERLOCK 6/	XF
6P4-9	MODIFIED SDC GND 6/	XF
6P4-10	CLOCK PULSE	0
6P4-11	MANUAL RANGE STROBE	I
6P4-12	MAX RANGE GATE	0
6P5-A	200VDC	I
6P5-B	200VDC GND	I
6P5-C	REPEATER BIT	I
6P5-D	AZIMUTH & EVALUATION GENERATOR BIT	I
6P5-E	DITHER	I
6P5-F	BCN BIT HI	I
6P5-G	BCN BIT LO	I
6P5-H	SHIELD GND	I

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TABLE VIII. SDC interface. (Continued)

Pin Number	Signal Nomenclature	Signal Interconnect 1/
6P5-J	POSITION ANALOG HI	I
6P5-K	POSITION ANALOG LO	I
6P5-L	SHIELD GND	I
6P5-M	COMPUTER/RECEIVER TEST FUNCTION	I
6P5-N	AFC TEST FUNCTION	I
6P5-P	TRANSMITTER TEST FUNCTION	I
6P5-R	ANTENNA SCAN TEST FUNCTION	I
6P5-S	TEST PULSE TEST FUNCTION	I
6P5-T	TURN RATE SAFE	I
6P5-U	FAIL OVERRIDE	0
6P5-V	28V TF TEST	0
6P5-W	ROLL SERVO MOTOR HI	I
6P5-X	ELEVATION SERVO MOTOR HI	I
6P5-Y	AZIMUTH SERVO MOTOR HI	I
6P5-Z	SHIELD GND	I
6P5-a	26VDC	I
6P5-b	28V IN EXPAND	0
6P5-c	CHASSIS GND	0
6P5-d	YAW RATE HI (OUT)	0
6P5-e	YAW RATE LO (OUT)	0
6P5-f	RADAR ALTITUDE HI (OUT	0
6P5-g	RADAR ALTITUDE LO (OUT)	0
6P5-h	CHASSIS GND	0
6P5-i	VELOCITY HI (OUT)	0
6P5-j	VELOCITY IO (OUT)	0
6P5-k	CHASSIS GND	0
6P5-m	CLIMB ANGLE HI (OUT)	0
6P5-n	CLIMB ANGLE LO (OUT)	0
6P5-p	CHASSIS GND	0
6P5-q	28V SIMULATION COMMAND (OUT)	0
6P5-r	COMPUTER UNBLANK	I
6P5-s	EOS	I
6P5-t	SHIELD GND	I
6P5-u	28V MODE OVERRIDE	I, 0
6P5-v	ANTENNA RECEIVER FAIL/SAFE	0
6P5-w	TEST PULSE FAIL/SAFE	0
6P5-x	COMPUTER FAIL/SAFE	0
6P5-Y	TRANSMITTER FAIL/SAFE	0
6P5-z	POWER SUPPLY FAIL/SAFE	0
6P5-AA	AIRCRAFT FAIL/SAFE	0
6P5-BB	BIT FAIL/SAFE	0
6P5-CC	FAIL/SAFE TO LAMP DRIVER	I
6P5-DD	SWEEP GENERATOR FAIL/SAFE	0
6P5-EE	28V MODE OVERRIDE & TA	I
6P5-FF	SERVO BIT	I
6P5-JJ	28V LOAD IN FAULT ISOLATION	I
6P5-KK	28V MODE OVERRIDE, LOAD & CSTA	I
6P5-LL	28V MODE OVERRIDE & CSTA	I

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TABLE VIII. SDC interface. (Continued)

Pin Number	Signal Nomenclature	Signal Interconnect 1/
6P5-MM	28V MODE OVERRIDE ANTENNA/RECEIVER	I
6P5-NN	28V TURN RATE SIMULATION	I
6P5-PP	28V CDCE OPERATE	I
6P6-A	WARNING FLAG HI	O
6P6-B	WARNING FLAG LO	O
6P6-C	IMS FAIL/SAFE HI	I
6P6-D	IMS FAIL/SAFE LO	I
6P6-E	RADAR ALTIMETER FAIL/SAFE HI	I
6P6-F	RADAR ALTIMETER FAIL/SAFE LO	I
6P6-G	NWDC FAIL/SAFE HI	I
6P6-H	NWDC FAIL/SAFE LO	I
6P6-J	RADAR ALTIMETER FAIL GND	O
6P6-K	SIDS ELEVATION HI	I
6P6-L	SIDS ELEVATION LO	I
6P6-M	SIDS AZIMUTH HI	I
6P6-N	SIDS AZIMUTH LO	I
6P6-P	SHIELD GND	I
6P6-R	VIDEO AMPLIFIER ENABLE GND	I
6P6-S	RANGE SWITCH NC	I
6P6-T	RANGE SWITCH EXCITATION	O
6P6-U	RANGE SWITCH NO	I
6P6-V	28V AGR TEST 1	I
6P6-W	28V AGR TEST 2	I
6P6-X	YAW RATE HI (IN)	I
6P6-Y	YAW RATE LO (IN)	I
6P6-Z	SHIELD GND	I
6P6-a	DATA WORD IN	I
6P6-b	DATA WORD IN BAR	I
6P6-c	IDENTIFICATION WORD IN	I
6P6-d	IDENTIFICATION WORD IN BAR	I
6P6-e	SHIELD GND	I
6P6-f	SYNCHRONIZATION IN	I
6P6-g	SYNCHRONIZATION IN BAR	I
6P6-h	SIGNAL RETURN	I
6P6-i	DATA READY IN	I
6P6-j	DATA READY IN BAR	I
6P6-k	SIGNAL RETURN	O
6P6-m	DATA WORD OUT	O
6P6-n	DATA WORD OUT BAR	O
6P6-p	IDENTIFICATION WORD OUT	O
6P6-q	IDENTIFICATION WORD OUT BAR	O
6P6-r	SYNCHRONIZATION OUT	O
6P6-s	SYNCHRONIZATION OUT BAR	O
6P6-t	DATA READY OUT	O
6P6-u	DATA READY OUT BAR	O
6P6-v	CHASSIS GND	O
6P6-z	SHIELD GND	I

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TABLE VIII. SDC interface. (Continued)

Pin Number	Signal Nomenclature	Signal Interconnect 1/
6P6-AA	ALTITUDE HI (IN)	I
6P6-BB	RADAR ALTITUDE LO (IN)	I
6P6-CC	VELOCITY HI (IN)	I
6P6-DD	VELOCITY LO (IN)	I
6P6-EE	SHIELD GND	I
6P6-FF	CLIMB ANGLE HI (IN)	I
6P6-GG	CLIMB ANGLE LO (IN)	I
6P6-HH	11.8VAC	I
6P6-JJ	26VAC PH C	I
6P6-KK	AC GND	I
6P6-LL	RANGE SET RETURN	0
6P6-NN	SIDS WRITE COMMAND HI	I
6P6-PP	SIDS WRITE COMMAND LO	I

1/ SIGNAL INTERCONNECT LEGEND: I = INPUT FROM AIRCRAFT SYSTEM OR FLR WRA
0 = OUTPUT TO AIRCRAFT SYSTEM OR FLR WRA
X = INTERCONNECT AMONG EDSCG WRA's
F = FORWARD CRTD
A = AFT CRTD

2/ NO INTERCONNECT REQUIRED FOR SINGLE CRTD (A-7D/E) EDSCG CONFIGURATIONS. APPLICABLE ONLY TO TA-7C/A-7K AIRCRAFT WIRING .

3/ FOR SINGLE CRTD (A-7D/E) EDSCG CONFIGURATION, SIGNAL INTERCONNECT = XF. FOR DUAL CRTD (TA-7C/A-7K) EDSCG CONFIGURATION, SIGNAL INTERCONNECT = 0.

4/ NOT CONNECTED IN AIRCRAFT WIRING. (IDENTIFICATION RESISTOR SENSE SIGNALS USED IN AUTOMATIC WRA TESTING ONLY).

5/ AIRCRAFT WIRING FOR THIS SIGNAL SHALL EXIST IN A-7D/E AIRCRAFT ONLY.

6/ COAX SHIELD UNUSED FOR THIS SIGNAL.

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TABLE IX. Forward CRTD interface.

Pin Number	Signal Nomenclature	Signal Interconnect 1/
8P1-1	28VDC	XS
8P1-2	28VDC GND	XS
8P1-3	+20VDC	I
8P1-4	RADAR SIGNAL GND	I
8P1-5	SELECT RADAR VIDEO	XS
8P1-7	SELECT RADAR VIDEO	I
8P1-8	MODIFIED SDC/IND INTERLOCK 2/	XS
8P1-10	FWD SELECT VIDEO	XS
8P1-11	MODIFIED IND/PSP INTERLOCK	I
8P1-12	28V NORMAL	XS
8P1-13	TV AVAILABLE GND	XS
8P1-14	28V IN TF + MEMORY TEST	XS
8P1-15	28V IN PPI	XS
8P1-16	28V EXPAND AVAILABLE	XS
8P1-17	28VDC SWITCHED	3/
8P1-18	28V RNG A	XS
8P1-19	28V RNG B	XS
8P1-20	28V RNG C	XS
8P1-21	28V RNG D	XS
8P1-22	28C RNG E	XS
8P1-23	SENSOR SELECT GND (FWD)	XS
8P1-24	TV AVAILABLE GND	I
8P1-25	FLIR AVAILABLE GND	I
8P1-27	AC GND FILTERED	XS
8P1-29	115VAC PH A FILTERED	XS
8P1-30	115VAC PH B FILTERED	XS
8P1-31	TEMPLATE VIDEO	I
8P1-33	MODIFIED SDC GND 2/	XS
8P1-34	MODIFIED SDC/IN INTERFLOCK 2/	XS
8P1-35	MODIFIED SDC GND 2/	XS
8P1-36	28V TV/SHRIKE LAMP EXCITATION	I
8P1-37	FAIL LAMP RETURN	I
8P1-38	CURSOR BRILL	XS
8P1-39	ADVISORY LAMP RETURN	I
8P1-40	200VDC	XS
8P1-41	200VDC GND	XS
8P1-42	28V HOLD IAMP	XS
8P1-44	SIDS SELECT GND	XS
8P1-45	5VAC RETURN	I
8P1-46	5VAC VARIABLE	I
8P1-47	115VAC PH C FILTERED	XS
8P1-48	SENSOR SELECT GND (AFT)	XS 4/
8P1-49	28V FAIL	XS
8P1-50	FLIR AVAILABLE GND	XS
8P1-51	SDC SIGNAL GND	XS
8P1-52	MANUAL RECEIVER GAIN	0
8P1-54	SHIELD GND	I
8P1-55	CHASSIS GND	I

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TABLE IX. Forward CRTD interface. (Continued)

Pin Number	Signal Nomenclature	Signal Interconnect 1/
8P1-56	SIDS AVAILABLE GND	I
8P1-58	AUTO-TILT AVAILABLE GND	XS
8P1-60	(IDA)	5/
8P1-62	TEMPLATE VIDEO	XS
8P1-65	AUTO-TILT COMMAND	O
8P1-66	AUTO-TILT LAMP	I
8P1-68	(IDB)	5/
8P1-69	(IDC)	5/
8P1-71	MODIFIED FWD/AFT INTERLOCK	XA 6/

1/ SIGNAL INTERCONNECT LEGEND: I = INPUT FROM AIRCRAFT SYSTEM OR FLR WRA
O = OUTPUT TO AIRCRAFT SYSTEM OR FLR WRA
X = INTERCONNECT AMONG EDSCG WRA's S = SDC
A = AFT CRTD

2/ COAX SHIELD UNUSED FOR THIS SIGNAL.

3/ FOR SINGLE CRTD (A-7D/E) EDSCG CONFIGURATION, SIGNAL INTERCONNECT = XS. FOR DUAL CRTD (TA-7C/A-7K) EDSCG CONFIGURATION, SIGNAL INTERCONNECT = I.

4/ AIRCRAFT WIRING FOR THIS SIGNAL SHALL EXIST IN A-7D/E AIRCRAFT ONLY.

5/ NOT CONNECTED IN AIRCRAFT WIRING. (IDENTIFICATION RESISTOR SENSE SIGNALS USED IN AUTOMATIC WRA TESTING ONLY.)

6/ NO INTERCONNECT REQUIRED FOR SINGLE CRTD (A-7D/E) EDSCG CONFIGURATIONS. APPLICABLE ONLY TO TA-7C/A-7K AIRCRAFT WIRING.

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TABLE X. Aft CRTD interface.

Pin Number	Signal Nomenclature	Signal Interconnect 1/
18P1-1	28VDC	XS
18P1-2	28VDC GND	XS
18P1-3	+20VDC	I
18P1-4	RADAR SIGNAL GND	I
18P1-10	AFT SELECT VIDEO	XS
18P1-12	28V NORMAL	XS
18P1-14	28V IN TF + MEMORY TEST	XS
18P1-15	28V IN PPI	XS
18P1-16	28V EXPAND AVAILABLE	XS
18P1-17	28VDC SWITCHED	I
18P1-18	28V RNG A	XS
18P1-19	28V RNG B	XS
18P1-20	28V RNG C	XS
18P1-21	28V RNG D	XS
18P1-22	28V RNG E	XS
18P1-23	SENSOR SELECT GND (AFT)	XS
18P1-24	TV AVAILABLE GND	I
18P1-25	FLIR AVAILABLE GND	I
18P1-27	AC GND FILTERED	XS
18P1-29	115VAC PH A FILTERED	XS
18P1-30	115VAC PH B FILTERED	XS
18P1-36	28V TV/SHRIKE LAMP EXCITATION	I
18P1-37	FAIL LAMP RETURN	I
18P1-38	CURSOR BRILL	XS
18P1-39	ADVISORY LAMP RETURN	I
18P1-40	200VDC	XS
18P1-41	200VDC GND	XS
18P1-42	28V HOLD LAMP	XS
18P1-44	SIDS SELECT GND	XS
18P1-45	5VAC RETURN	I
18P1-46	5VAC VARIABIE	I
18P1-47	115VAC PH C FILTERED	XS
18P1-49	28V FAIL	XS
18P1-51	SDC SIGNAL GND	XS
18P1-52	MANUAL RECEIVER GAIN	0
18P1-54	SHIELD GND	0
18P1-55	CHASSIS GND	I
18P1-56	SIDS AVAILABLE GND	I
18P1-58	AUTO-TILT AVAILABLE GND	XS
18P1-60	(IDA)	2/
18P1-65	AUTO-TILT COMMAND	0
18P1-66	AUTO-TILT LAMP	I
18P1-68	(IDB)	2/
18P1-69	(IDC)	2/
18P1-70	MODIFIED FWD/AFT INTERLOCK	XF

1/ SIGNAL INTERCONNECT LEGEND: I = INPUT FROM AIRCRAFT SYSTEM OR FLR WRA
0 = OUTPUT TO AIRCRAFT SYSTEM OR FLR WRA
X = INTERCONNECT AMONG EDSCG WRA's
F = FORWARD CRTD
S = SDC

2/ NOT CONNECTED IN AIRCRAFT WIRING. (IDENTIFICATION RESISTOR SENSE SIGNALS USED IN AUTOMATIC WRA TESTING ONLY.)

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TABLE XI. Aircraft sensor priorities. 1/

A/C SENSOR CMD			FWD CRTD 2/		AFT CRTD		FWD	AFT	XMTR	
SIDS	TV	FLIR	NORM/OVRD	DISPLAY	NORM/OVRD	DISPLAY	LAMP 2/	LAMP	SILENCE	
			NORM	RDR	NORM	RDR	---	---		
		X	NORM	FLIR	NORM	FLIR	TV	TV	3/	4/
	X		NORM	TV	NORM	TV	TV	TV	3/	4/
	X	X	NORM	TV	NORM	TV	TV	TV	3/	4/
X			NORM	SIDS	NORM	SIDS	SIDS	SIDS	3/	4/
X		X	NORM	SIDS	NORM	SIDS	SIDS, TV	SIDS, TV	3/	4/
X	X		NORM	SIDS	NORM	SIDS	SIDS, TV	SIDS, TV	3/	4/
X	X	X	NORM	SIDS	NORM	SIDS	SIDS, TV	SIDS, TV	3/	4/
—				OVRD	RDR	NORM	RDR	---	---	
		X	OVRD	RDR	NORM	FLIR	TV	TV		
	X		OVRD	RDR	NORM	TV	TV	TV		
	X	X	OVRD	RDR	NORM	TV	TV	TV		
X			OVRD	RDR	NORM	RDR	SIDS	SIDS		
X		X	OVRD	RDR	NORM	FLIR	SIDS, TV	SIDS, TV		
X	X		OVRD	RDR	NORM	TV	SIDS, TV	SIDS, TV		
X	X	X	OVRD	RDR	NORM	TV	SIDS, TV	SIDS, TV		
			NORM	RDR	OVRD	RDR	---	---		
		X	NORM	FLIR	OVRD	RDR	TV	TV	3/	
	X		NORM	TV	OVRD	RDR	TV	TV	3/	
	X	X	NORM	TV	OVRD	RDR	TV	TV	3/	
X			NORM	SIDS	OVRD	SIDS	SIDS	SIDS	3/	4/
X		X	NORM	SIDS	OVRD	SIDS	SIDS, TV	SIDS, TV	3/	4/
X	X		NORM	SIDS	OVRD	SIDS	SIDS, TV	SIDS, TV	3/	4/
X	X	X	NORM	SIDS	OVRD	SIDS	SIDS, TV	SIDS, TV	3/	4/
			OVRD	RDR	OVRD	RDR	---	---		
		X	OVRD	RDR	OVRD	RDR	TV	TV		
	X		OVRD	RDR	OVRD	RDR	TV	TV		
	X	X	OVRD	RDR	OVRD	RDR	TV	TV		
X			OVRD	RDR	OVRD	RDR	SIDS	SIDS		
X		X	OVRD	RDR	OVRD	RDR	SIDS, TV	SIDS, TV		
X	X		OVRD	RDR	OVRD	RDR	SIDS, TV	SIDS, TV		
X	X	X	OVRD	RDR	OVRD	RDR	SIDS, TV	SIDS, TV		

1/ Table describes operation for dual CRTD (TA-7C or A-7K) EDSCG subsystem configuration

2/ "FWD CRTD" and "FWD LAMP" columns describe operation for a single CRTD (A-7D or A-7E) EDSCG subsystem configuration.

3/ Automatic transmitter silencing shall occur for these sensor/CRTD conditions in single CRTD EDSCGs,

4/ Automatic transmitter silencing shall occur for these sensor/CRTD conditions in dual CRTD EDSCGs.

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TABLE XII. AGR tracking accuracy for clutter targets.

Reflectivity	Grazing Angle (Degrees)	Grazing Angle (Radians)	Slant Range for ±25 foot (±7.6 m) Accuracy (1 σ)	Slant Range For Accuracy of ± .25% Range to Target (1 σ)
-20dB	4	0.07	1000 to 10,000 ft (305 to 3050 m)	10,000 to 22,000 feet (3050 to 6710 m)
-20dB	6	0.10	1000 to 10,000 feet (305 to 3050 m)	10,000 to 35,000 feet 3050 to 10,700 m)
-20dB	8	0.14	1000 to 10,000 feet (305 to 3050 m)	10,000 to 45,000 feet 3050 to 13,700 m)
-20dB	10 to 70	0.17 to 1.22	1000 to 10,000 feet (305 to 3050 m)	10,000 to 60,000 feet (3050 to 18,300 m)

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TABLE XIII. CDCE data channel information bit significance.

EDSCG DISPLAY MODE	DATA WORD TYPE	ID WORD CODE	DATA WORD BIT SPECIFICATION
CDCE	Slant Range of the range cursor	Bit 14-Digital one Bit 15-Digital zero	Bit 5 (LSB) = 30 feet. (9.1 m) Bit 18 (MSB) = 245,760 feet (74,908 m) The sign of the word is assumed to be positive.
EXPAND	Slant Range of the range cursor	Bit 14-Digital one Bit 15-Digital zero	Bit 5 (LSB) = 30 feet. (9.1 m) Bit 18 (MSB) = 245,760 feet (74,908 m) The sign of the word is assumed to be positive.
CDCE	Sine of the angle of the azimuth cursor	Bit 14-Digital one Bit 15-Digital zero	Bit 8 (LSB) = 2^{-10} Bit 17 (MSB) = 0.5. Bit 18 (Sign) - Positive is digital zero. Bits 3 through 7 will be present. Negative numbers shall be represented in sign bit magnitude form.
GND BCN or GND MAP	Altitude	Bit 14-Digital zero Bit 15-Digital zero	Bit 5 (LSB) = 30 feet. (9.1 m) Bit 18 (MSB) = 30,720 feet (9363 m) The sign of the word is assumed to be positive.

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TABLE XIV. Protection interface.

FROM OUTPUT:	TO INPUT:	SIGNAL NAME	SIGNAL LEVEL
(IP3-AA) W1P3-u	6P1-n	Modified PSP Command	+28 V (Current-Limited)
6P4-4 */ 6P4-8 */	8P1-8 */ 8P1-34 */	Modified SCD/IND Interlock	+28 V (Current-Limited) (From SDC) +20 VDC (From CRTD)
6P4-9 6P4-7	8P1-35 8P1-33	Modified SDC GND	28 V GND
8P1-11 */	(1P3-w) */ W1P3-j	Modified IND/PSP Interlock	+28 V (Current-Limited) (From CRTD) +20 VDC (From PSP)
8P1-71	18P1-70	Modified FWD/AFT Interlock	+20 VDC

*/ Signals are both driven from and received at these interfaces at different times during radar time-out.

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TABLE XV. Protected signals.

FROM OUTPUT:	TO INPUT:	SIGNAL NAME	SIGNAL LEVEL
6P1-F */	(1P3-u) W1P4-F	Radar Silence GND	28 V GND (or OPEN)
6P1-G */	(1P3-i) WP14-G	IMS Fail GND	28 V GND (or OPEN)
6P1-BB */	8P1-42	28 V Hold Lamp (Fwd)	28 V (or OPEN)
6P2-N	8P1-58*/	Auto-Tilt Available GND	V GND (or OPEN)
6P5-b */	(1P6-K) W1P4-KK	28 V in Expand	28 V GND (or 28 V Pull-Up)
8P1-48 */	6P2-MM	Sensor Select GND (Aft)	28 V GND
8P1-65 */ 18P1-65 */	(1P3-L) W1P3-Y	Auto-Tilt Command	28 V GND (or OPEN)
(1P3-U) W1P3-M	8P1-66 */ 18P1-66 */	Auto-Tilt Lamp	28 V (or OPEN)

*/ Protection results from signal interrupt inside WRA connected at this interface.

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TABLE XVI. SDC interface connectors.

Reference Designator	Receptacle Type	Function
6J1	Bendix JTPOORE-24-61P	Power and signals
6J2	Bendix JTPOORE-24-61PA	Power and signals
6J3	Deutsch CVC6063E-61-21P-1	Signals
6J4	Deutsch CVC6063E-61-21PW	Signals
6J5	Bendix JTPOORE-24-61-PB	Power and signals
6J6	Bendix JTPOORE-24-61-PC	Power and signals
6J7	Bendix 33506-2	Test signal

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TABLE XVII. SDC vibration test.

VIBRATION TEST	FREQUENCY RANGE (Hz)	MAXIMUM AMPLITUDE	APPLICABLE FIGURE
Random (Operational)	20-2000	$.03g^2/Hz$	1
Random (Endurance)	20-2000	$.14g^2/HZ$	2
High Fire Rate Pulse			
X - axis	100-2000	15g-peak	3
Y,Z - axes	100-2000	15g-peak	4
Low Fire Rate Pulse			
X - axis	67-2000	15g-peak	5
Y,Z - axes	65-2000	15g-peak	6

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TABLE XVIII. CRTD vibration test.

VIBRATION TEST	FREQUENCY RANGE (Hz)	MAXIMUM AMPLITUDE	APPLICABLE FIGURE
Random (Operational)	20-2000	$.005g^2/Hz$	7
Sinusoidal	5-500	2g-peak	8
High Fire Rate Pulse	100-2000	1.5g-peak	9
Low Fire Rate Pulse	67-2000	2.5g-peak	10

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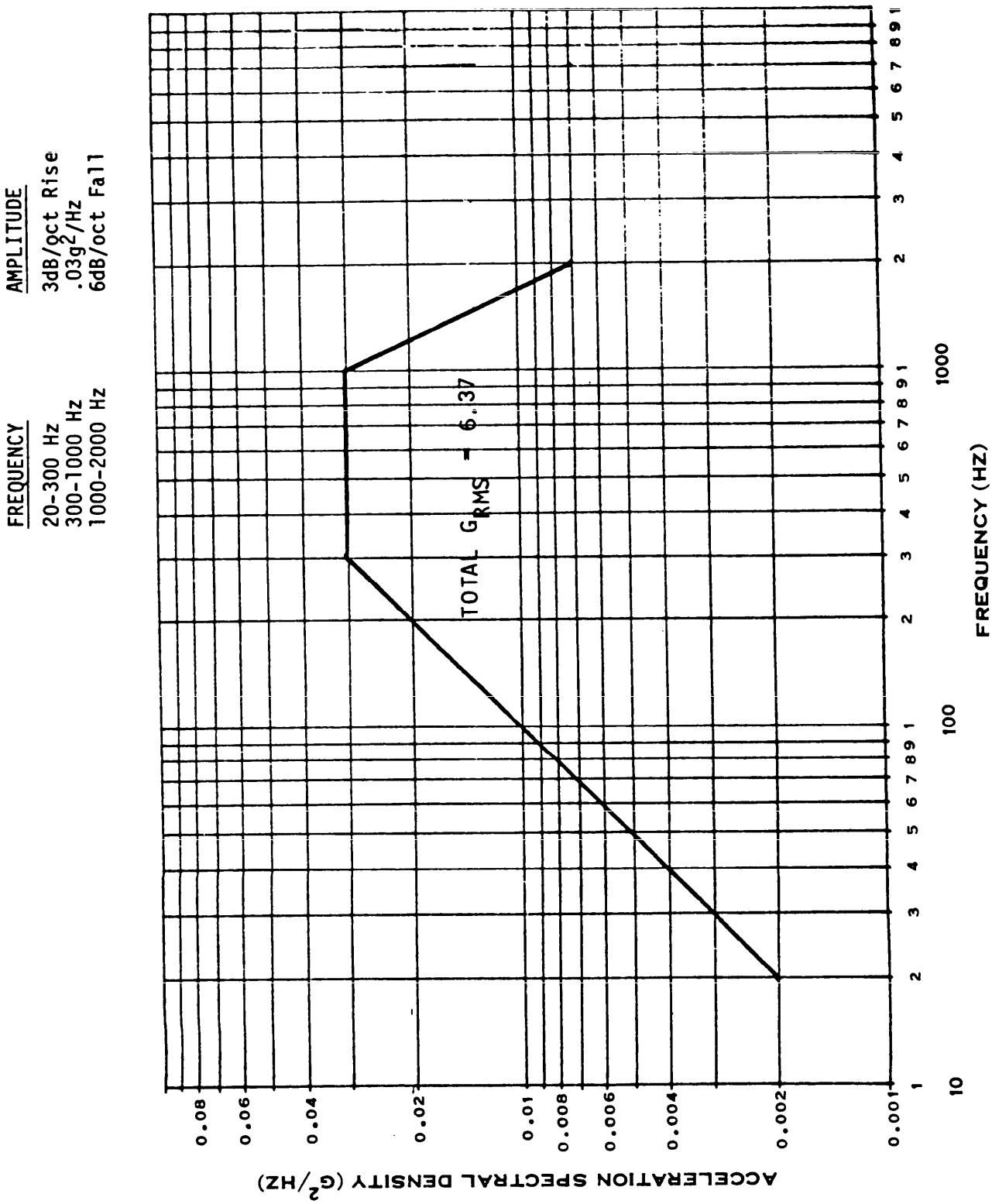


FIGURE 1 - SDC vibration test spectrum - random (operational).

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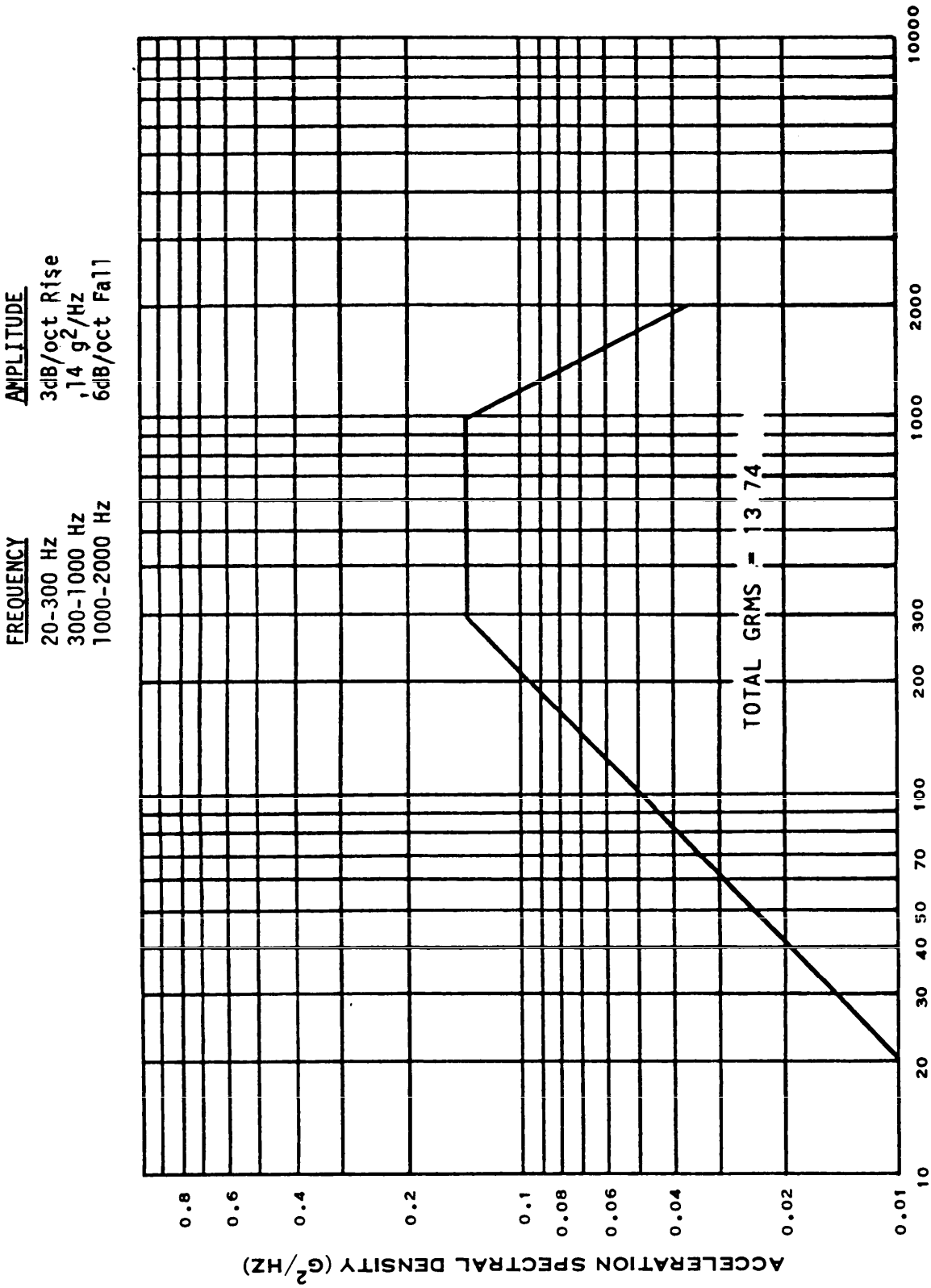


FIGURE 2 - SDC vibration test spectrum - random (endurance).

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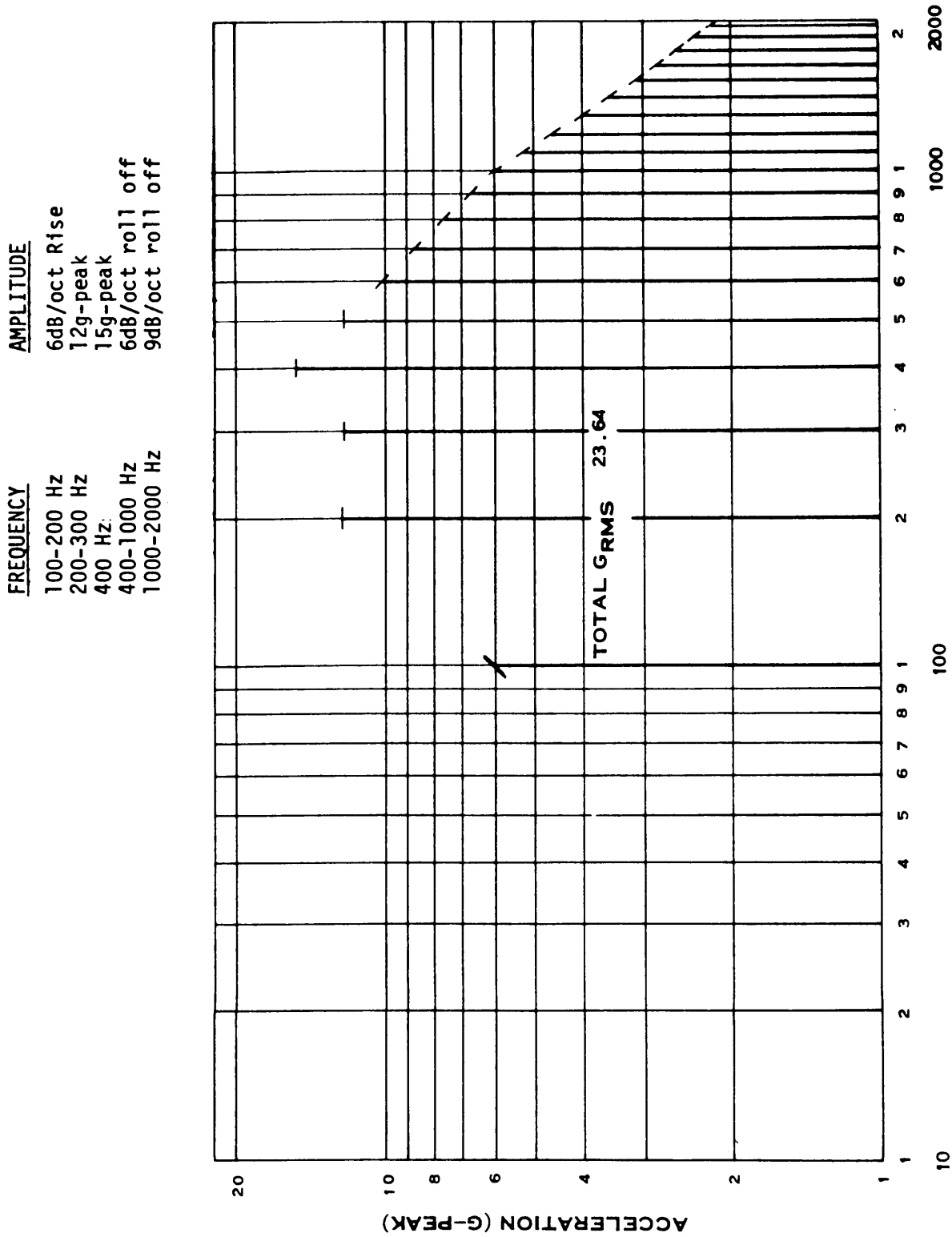


FIGURE 3 - SDC vibration test spectrum - high fire rate pulse (X-axis).

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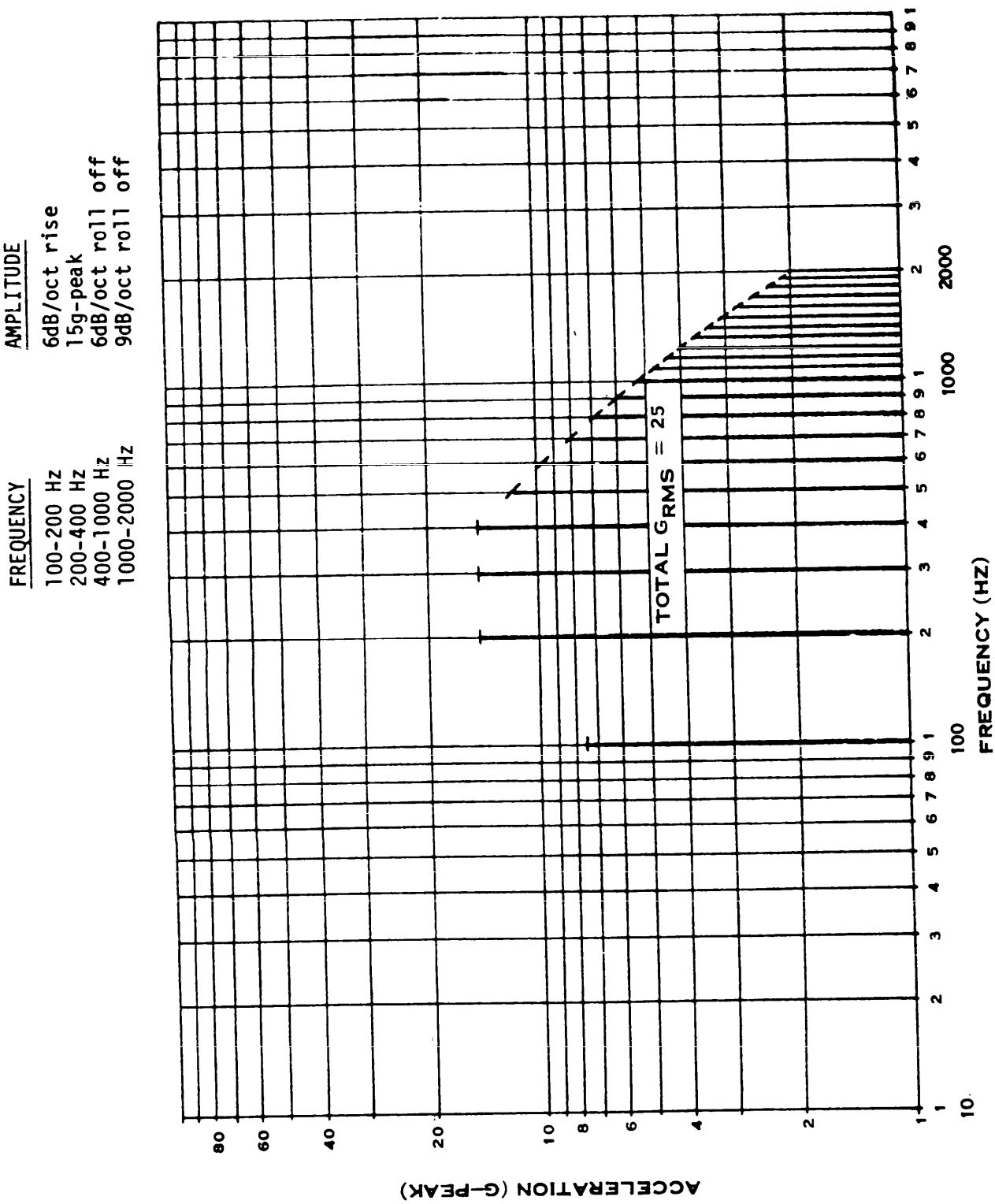


FIGURE 4 - SDC vibration test spectrum - high fire rate pulse (Y,Z axes).

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Check the source to verify that this is the current version before use.

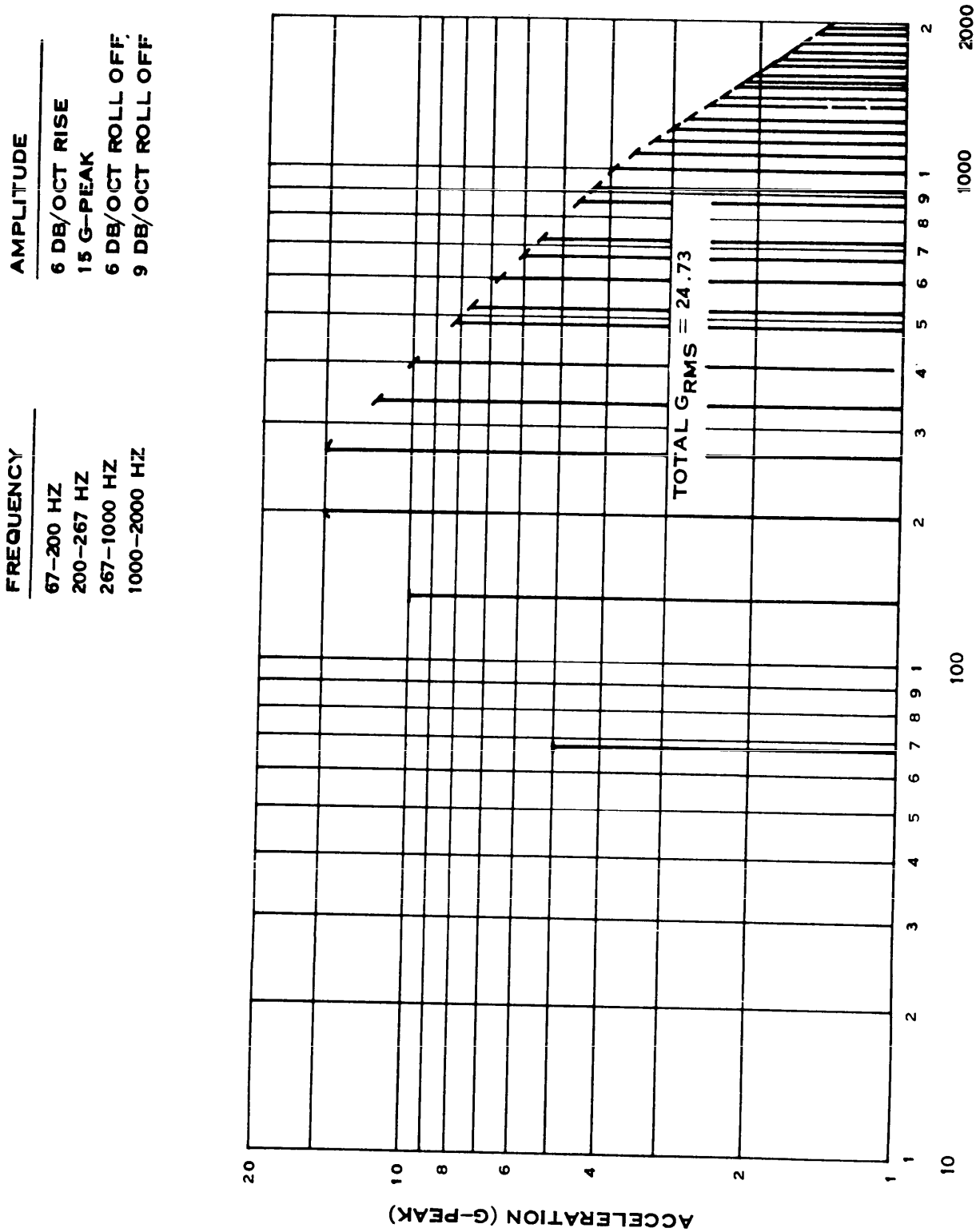


FIGURE 5 – SDC vibration test spectrum - low fire rate pulse (X-axis).

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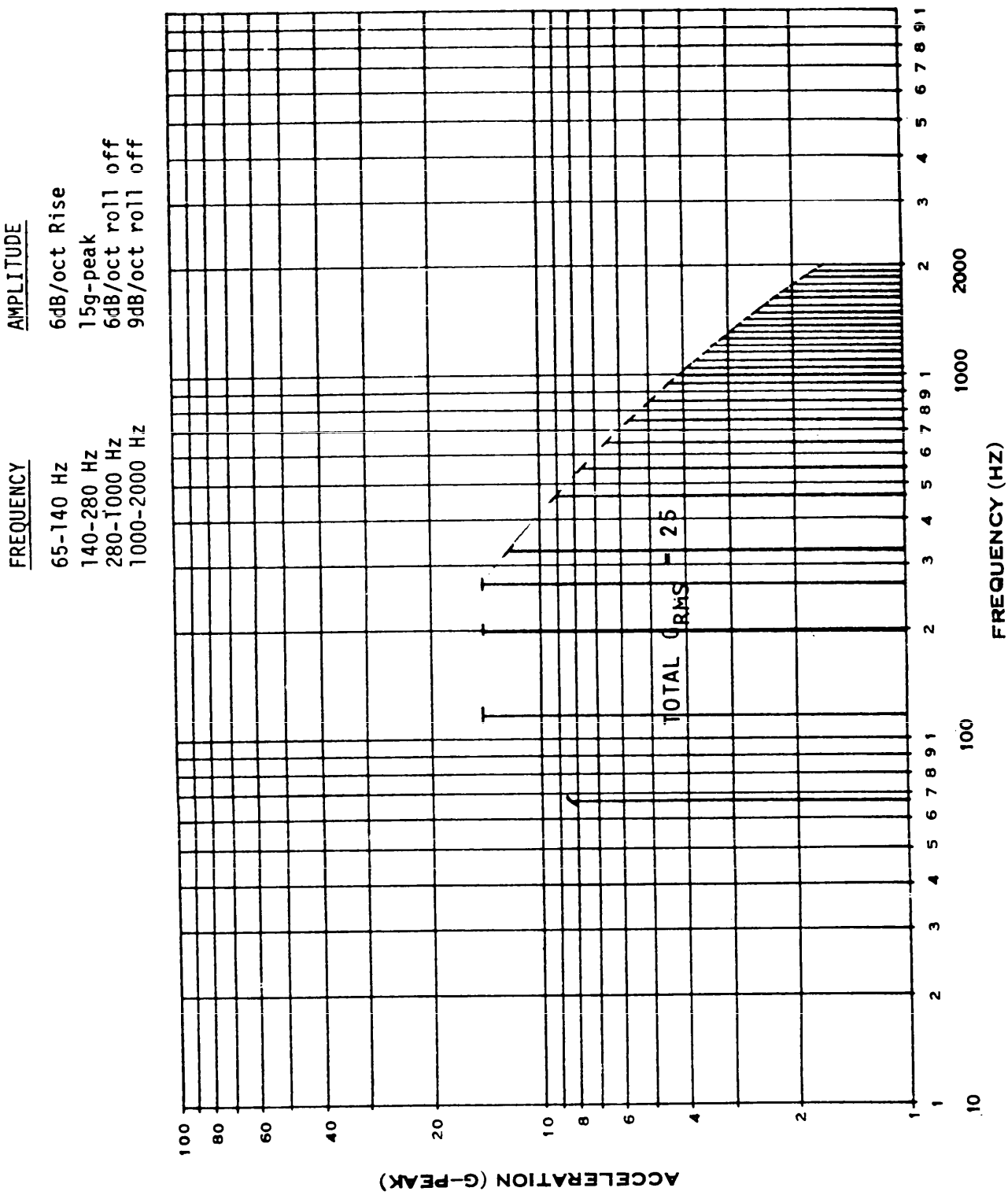


FIGURE 6 - SDC vibration test spectrum - low fire rate pulse (Y, Z axes).

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Check the source to verify that this is the current version before use.

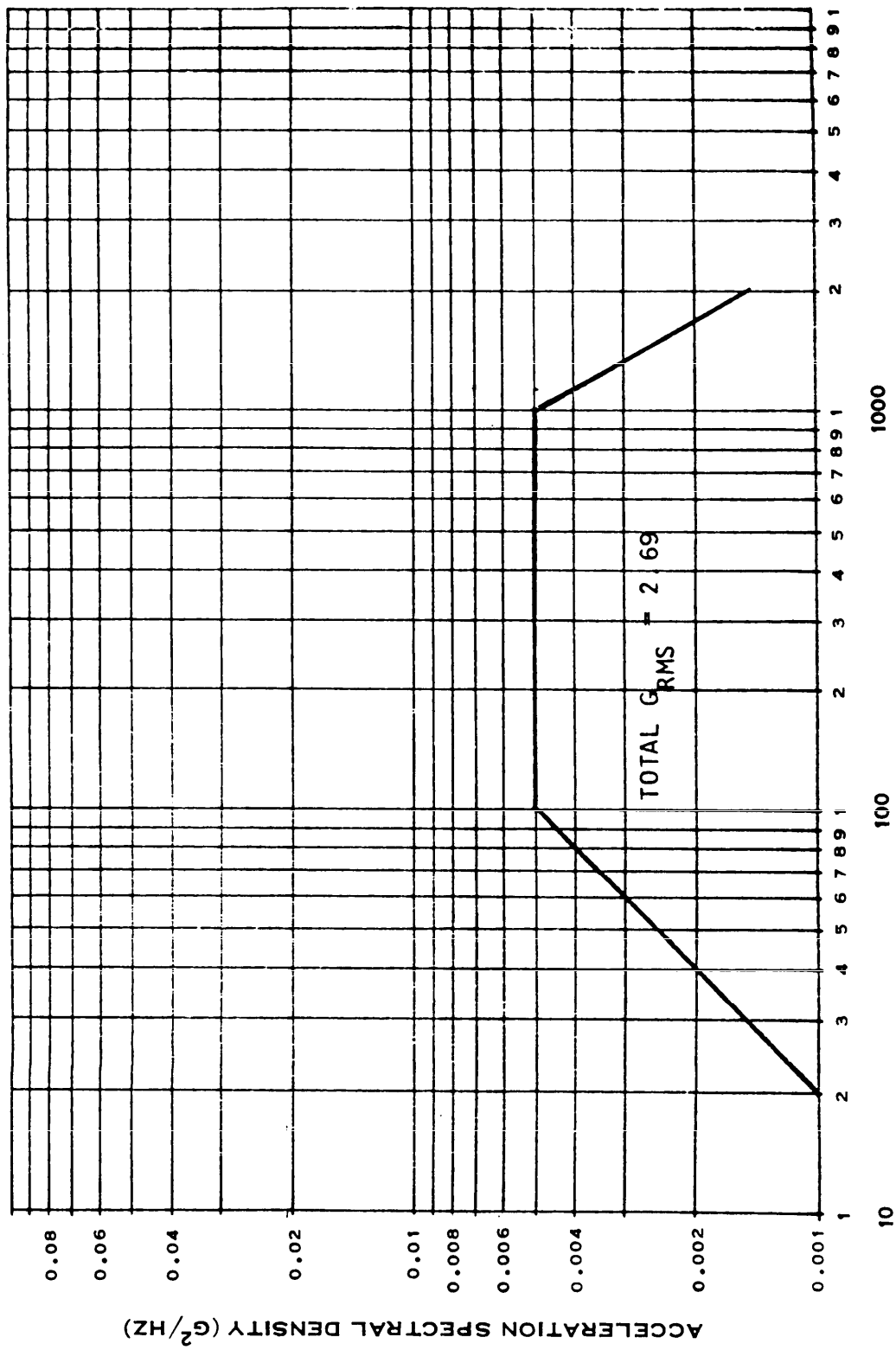


FIGURE 7 - CRTD vibration test spectrum - random (operational).

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Check the source to verify that this is the current version before use.

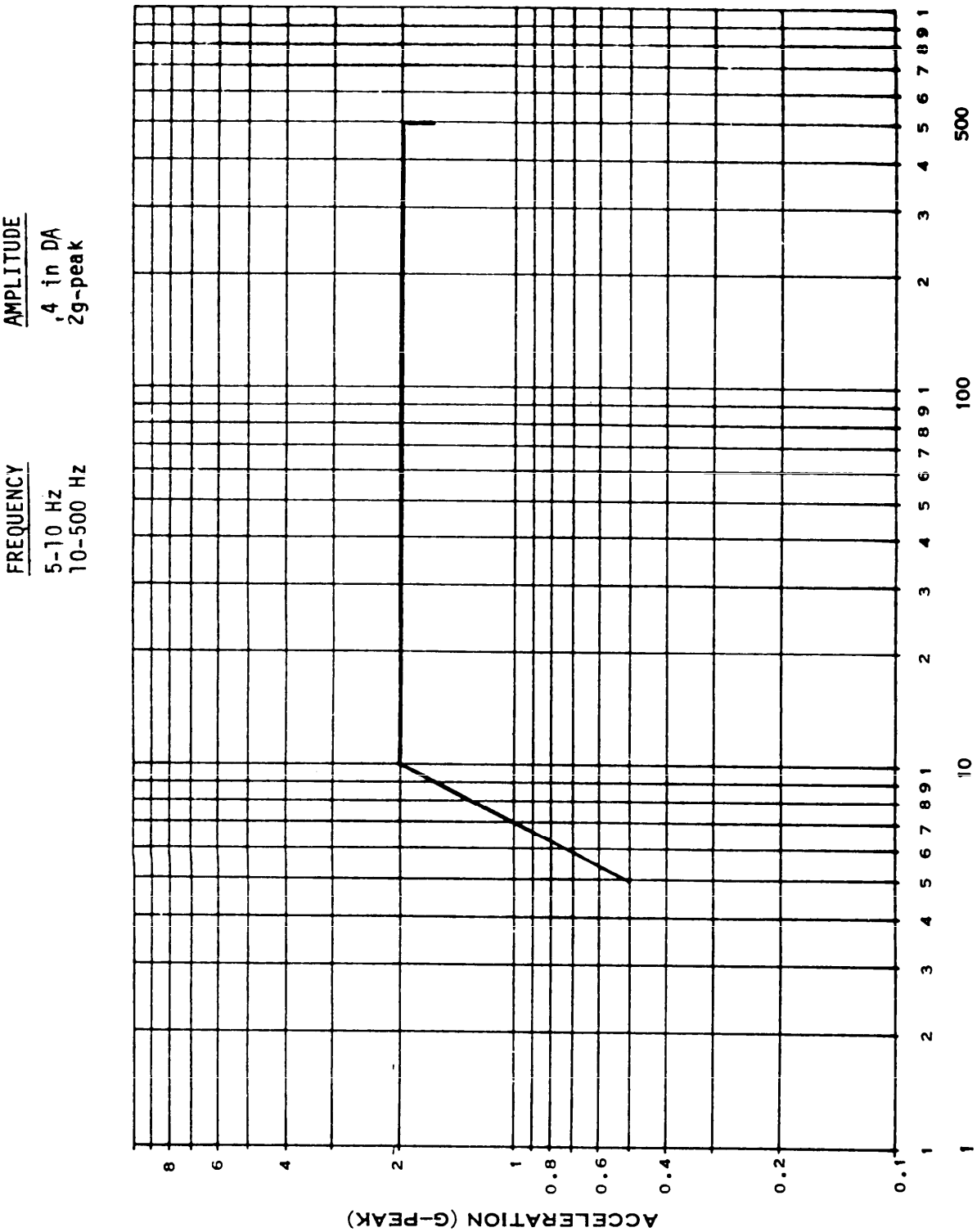


FIGURE 8 - CRID vibration test spectrum - sinusoidal.

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Check the source to verify that this is the current version before use.

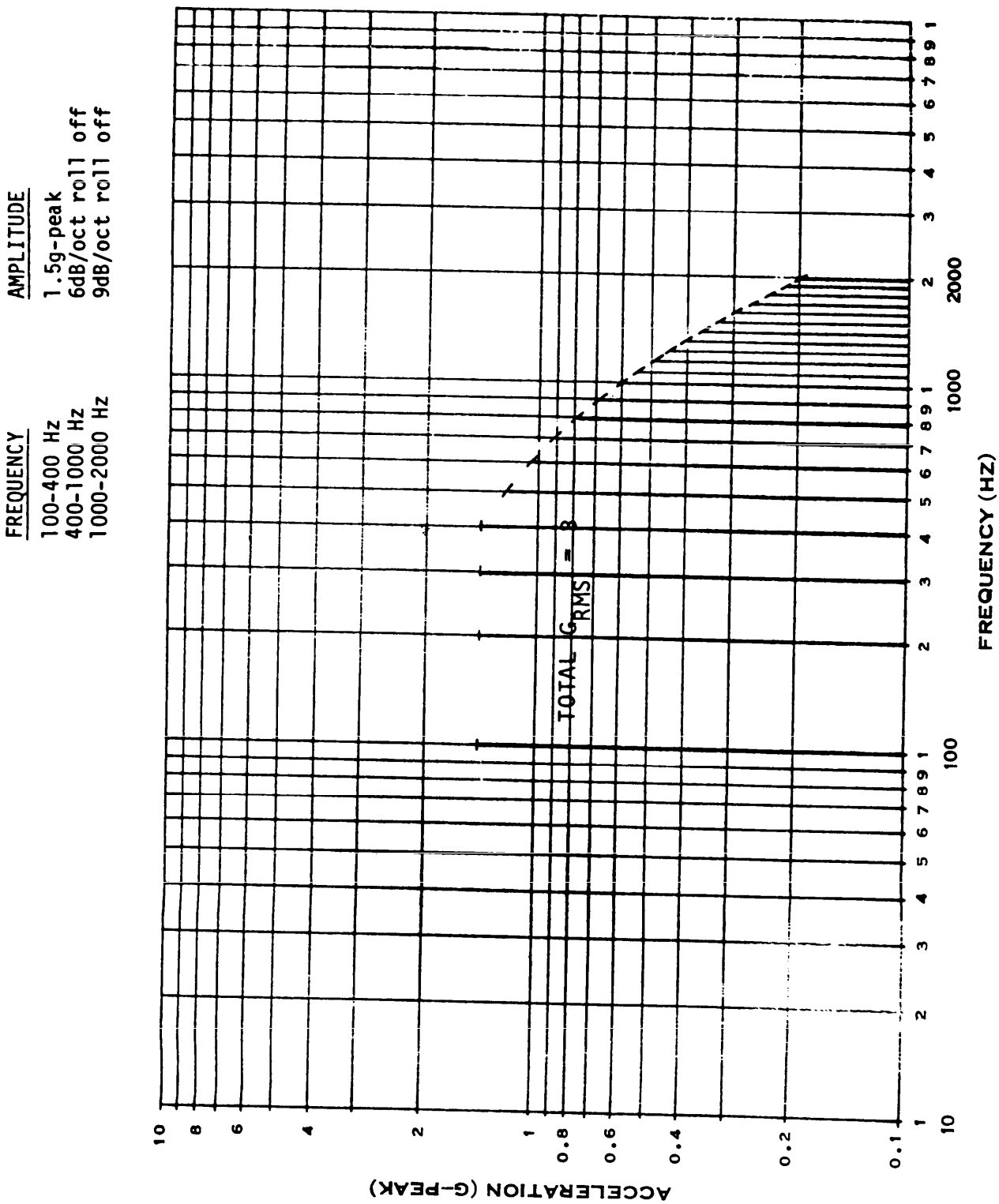


FIGURE 9 - CRTD vibration test spectrum - high fire rate pulse.

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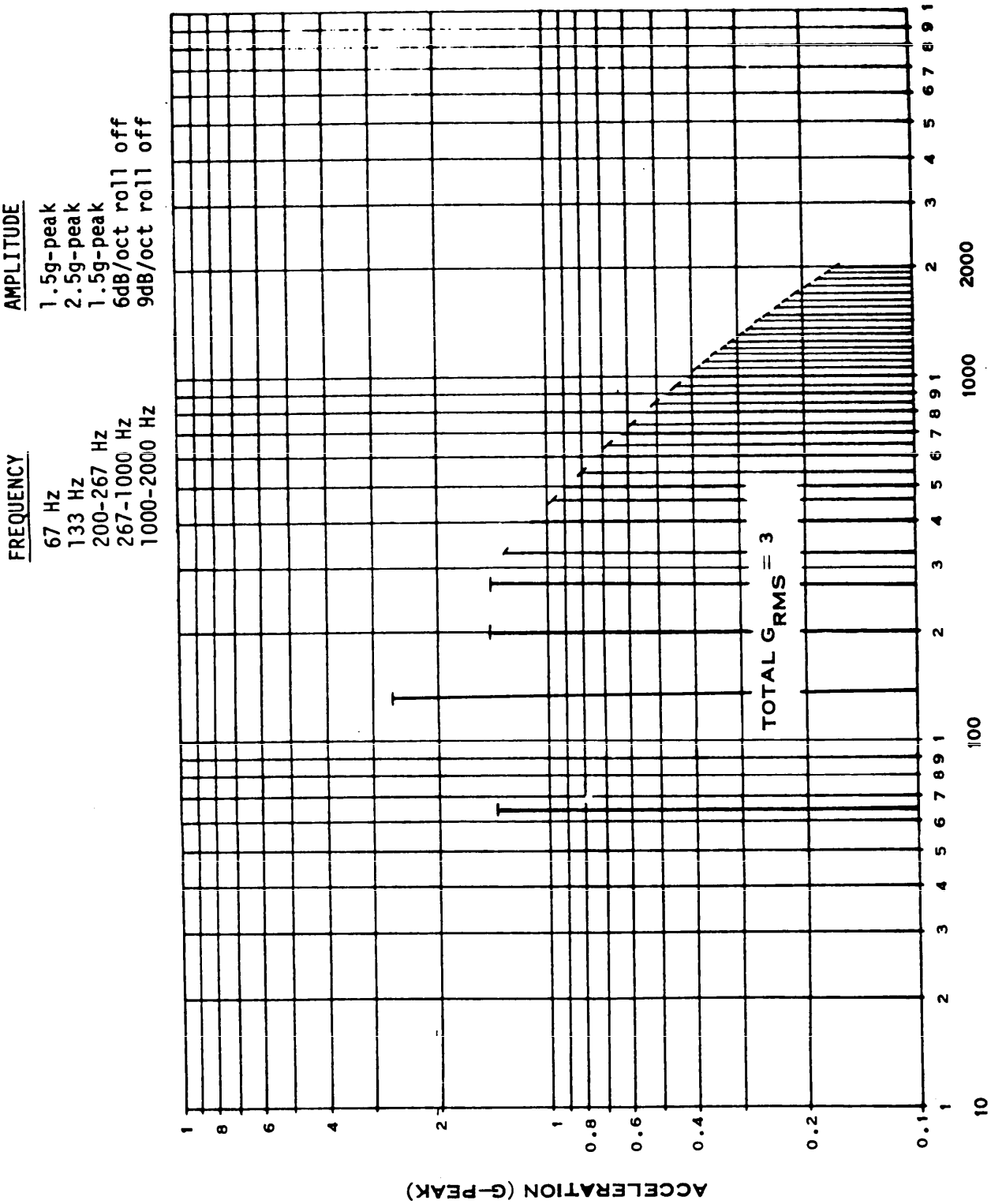
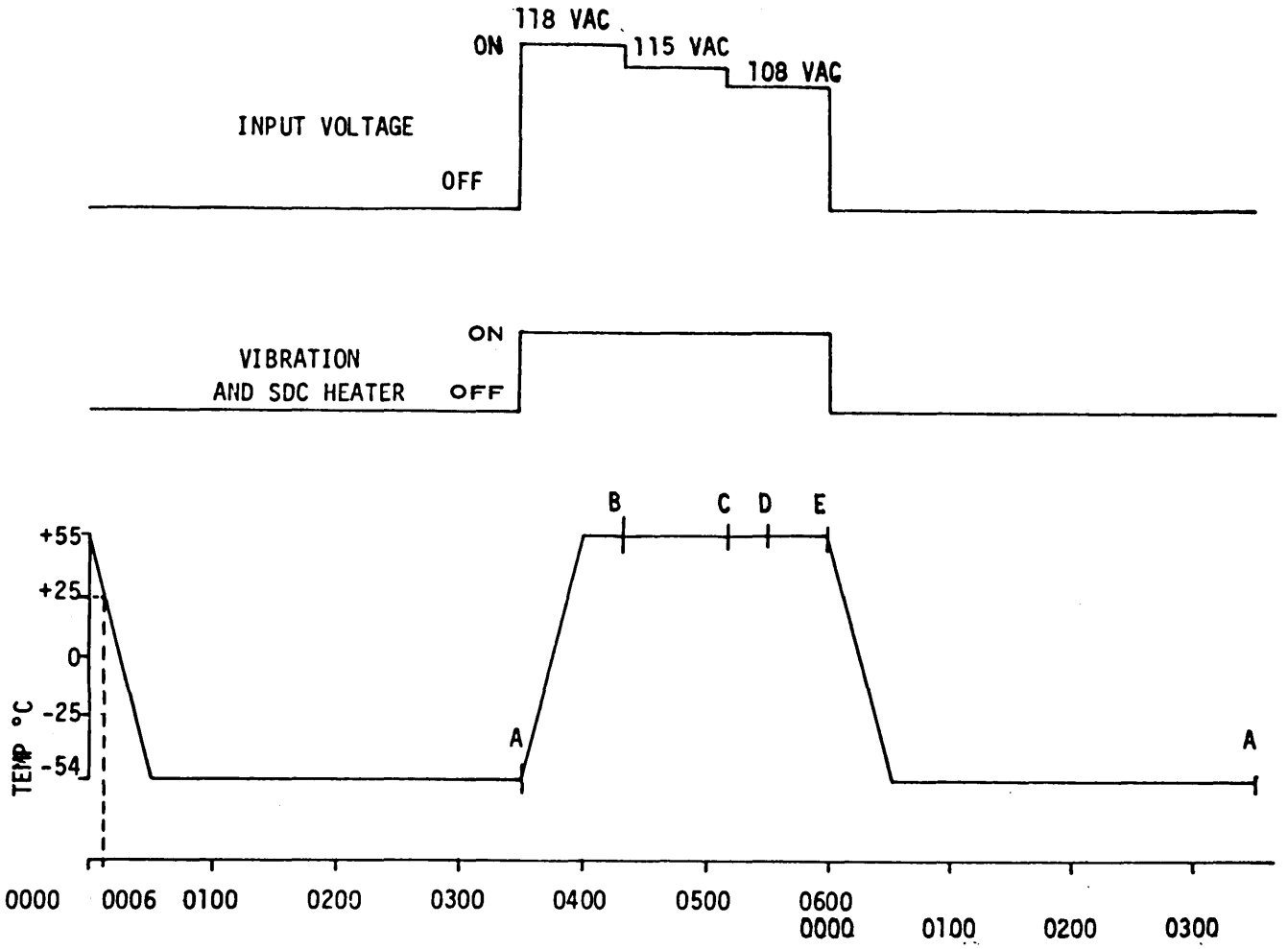


FIGURE 10 - CRTD vibration test spectrum - low fire rate pulse.

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0000 PROGRAM CHAMBER TO -54°

0006 PROGRAM CHAMBER TO -54°C (FIRST CYCLE ONLY)

A. 0330 ADJUST CHAMBER TEMPERATURE FOR +55°C EQUIPMENT ON, VIBRATION ON, ADJUST INPUT VOLTAGE FOR 118 VAC. RUN COLD PERFORMANCE TEST. LEAVE EQUIPMENT ON.

B. 0420 ADJUST INPUT VOLTAGE FOR 115 VAC.

C. 0510 ADJUST INPUT VOLTAGE FOR 108 VAC.

D. 0530 RUN HOT PERFORMANCE TEST.

E. 0600 EQUIPMENT OFF, VIBRATION OFF. PROGRAM CHAMBER TO -54°C.

FIGURE 11 - Environmental burn-in test cycle.

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